

Money and Success in Football

Does Financial Strength Predict League Position? *Sports Economics Seminar — Summer Semester 2026*

Md. Muktadir

July 11, 2026

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1 Executive Summary

Scale of the study: 976 team-seasons across 5 major European leagues (Bundesliga, La Liga, Ligue 1, Premier League, Serie A) covering 10 seasons (2015–16 to 2024–25). All leagues and seasons are included in the analysis.

Core finding: The Pearson correlation between Financial Rank and final league position is $r = 0.813$ (Spearman $\rho = 0.814$, both $p < 0.001$). The best-fitting model — Model H, which adds a quadratic (squared) term to Financial Rank — explains $R^2 = 0.671$ of all variation in final standings. In plain terms, knowing only how wealthy a club is relative to its peers allows us to correctly explain **67%** of where they finish.

Champions by financial rank: 76% of the 50 league titles in the study were won by the squad with the **highest market value** in their league that season (Financial Rank = 1). Only 4 champions came from outside the financial top three — and only one of them (Leicester City 2015–16) qualifies as a genuine statistical outlier.

The one true exception: Leicester City 2015–16 finished Financial Rank 11 in the Premier League yet won the title. The model predicted they would finish 11th; instead they came 1st. Their residual of **-10.5 positions** is the largest shock in the entire dataset and a once-in-a-generation event.

Extended robustness (Step 7): the core finding is additionally stress-tested against a two-period time split, an alternative outcome variable (points instead of position), and a cross-sport comparison with the NBA’s salary-capped league. It survives all three — though the NBA comparison shows that a hard salary cap can meaningfully weaken, without eliminating, the link between spending and winning.

2 Introduction

2.1 Why Does This Question Matter?

The five major European football leagues — the English Premier League, Spanish La Liga, German Bundesliga, Italian Serie A, and French Ligue 1 — collectively generate over €18 billion in annual revenue, with squad transfer spending exceeding €12 billion in recent seasons. With this much money flowing through the sport, a natural and important question arises: **does financial investment actually buy success, or can a well-organised, under-funded club systematically compete at the top level?**

This question is not just about football. It has direct implications for:

- **Competitive balance policy** — UEFA introduced Financial Fair Play precisely to prevent the wealthiest clubs from buying their way to dominance. If money truly determines outcomes, these policies become even more critical.
- **Fan engagement** — If leagues are financially predetermined, the drama of competition is diminished for supporters of smaller clubs.
- **Sports economics theory** — Understanding how resource endowments translate to tournament outcomes informs models across all competitive sports.

2.2 What We Set Out to Answer

This study addresses three interrelated questions using ten years of data from all five top-tier European leagues:

1. **Direction and strength:** Is there a statistically significant relationship between a squad’s financial rank within its league and its final position? How strong is that relationship?
2. **Model form:** Does a simple linear relationship capture the data, or is there a more complex (e.g., non-linear) pattern? Does the relationship differ across leagues or change over time?
3. **Exceptions:** Which clubs defy the financial hierarchy, and how statistically unusual are they? Is Leicester City 2015–16 a miracle or merely a rounding error?

2.3 What This Study Adds

Previous research (Szymanski & Kuypers, 1999; Frick, 2007) has shown that wage bills and transfer spending predict league success. This study makes two additional contributions. First, instead of raw market values, we use a **within-league financial rank** — which club was the n -th wealthiest in their specific league that season. This removes the scale problem: a €300M squad is the richest in Ligue 1 but the fifth-richest in the Premier League. By ranking within leagues, we can pool all five leagues and make fair comparisons. Second, we introduce a **residual-based outlier framework** to systematically identify and quantify clubs that dramatically beat or missed their financial prediction.

3 The Data We Used

3.1 Where the Data Comes From

Our dataset was assembled by combining two sources:

- **ESPN FC** provided final league standings (League Position) for all five leagues across the 2015–16 through 2024–25 seasons.
- **Transfermarkt** provided squad market values — the estimated total transfer market worth of all players in each squad — at the start of each season.

After merging these two sources and removing any rows with missing values on key variables, we have a clean dataset of 976 team-season observations. Every team that played a full season in one of the five leagues during the study period appears once per season.

Table 1: Sample size and position/market-value statistics by league.

Group	N	Mean Pos.	SD Pos.	Mean MV (M EUR)	SD MV (M EUR)
ALL LEAGUES (combined)	976	10.28	5.66	277.7	254.3
Bundesliga	180	9.50	5.20	237.2	205.6
La Liga	200	10.50	5.78	256.7	269.4
Ligue 1	196	10.32	5.69	184.7	193.9
Premier League	200	10.50	5.78	458.0	303.5
Serie A	200	10.50	5.78	245.8	182.4

Table 2: Correlation between Financial Rank and final League Position by league.

Group	N	Pearson r (FR vs Pos.)	Spearman rho (FR vs Pos.)
ALL LEAGUES (combined)	976	0.813	0.814
Bundesliga	180	0.783	0.783
La Liga	200	0.787	0.787
Ligue 1	196	0.791	0.791
Premier League	200	0.801	0.801
Serie A	200	0.890	0.890

3.2 Descriptive Statistics

The table below summarises the key characteristics of the dataset, broken down by league and combined.

Reading Tables 1a and 1b: The “Combined” row (highlighted in blue) captures all 976 observations. Table 1a shows that market values vary substantially across leagues — the Premier League average squad value is considerably higher than Ligue 1, for example — but positions are naturally centred around mid-table in all leagues. Table 1b shows the key finding from raw data alone: the Pearson correlation between Financial Rank and final position is above 0.80 in every single league, and Serie A reaches 0.89. These numbers are striking — without running any model, the raw correlations already tell us money is a very powerful predictor of where clubs finish.

3.3 Variable Definitions

The table below defines each variable used in the analysis.

The key design choice — Financial Rank instead of raw market value: Using Financial Rank (FR) rather than the raw €-value is crucial for a multi-league analysis. Consider: a squad worth €300M is the *richest* in Ligue 1 but only the *fifth-richest* in the Premier League. If we used raw €-values and pooled all leagues, we would be comparing apples to oranges. By ranking each squad within its own league-season, we create a fair, scale-free measure. FR = 1 always means “the wealthiest club in this league this season,” regardless of whether that squad is worth €600M or €150M.

3.4 How Financial Rank Is Distributed

The chart below shows the distribution of Financial Rank values across all five leagues. Since every league-season has exactly as many clubs as positions, the distribution should be near-uniform — and indeed it is. This confirms that the ranking system is clean and well-balanced across the dataset.

Table 3: Variable definitions.

Variable	Type	Range	Description
League_Position	Integer	1–18/20	Final league standing; 1 = champion, higher number = worse finish
Financial_Rank (FR)	Integer	1–18/20	Squad rank by market value within league-season; 1 = richest squad
Market_Value_M_EUR	Numeric	10–3300+	Total squad market value in millions of euros (from Transfermarkt)
Normalized_Value	Numeric	0–1	Market value normalised within league-season (min-max or value share)
Log_Normalized_Valu	Numeric	$-\infty$ –0	Natural log of Normalized_Value; makes the skewed distribution more symmetric
Champion_bin	Binary (0/1)	0 or 1	1 if the team won the league title that season, 0 otherwise
Season_Index	Integer	1–10	Numeric season counter: 1 = 2015–16, 10 = 2024–25

Distribution of Financial Rank by League

N = 976 team-seasons; values near-uniform within each league

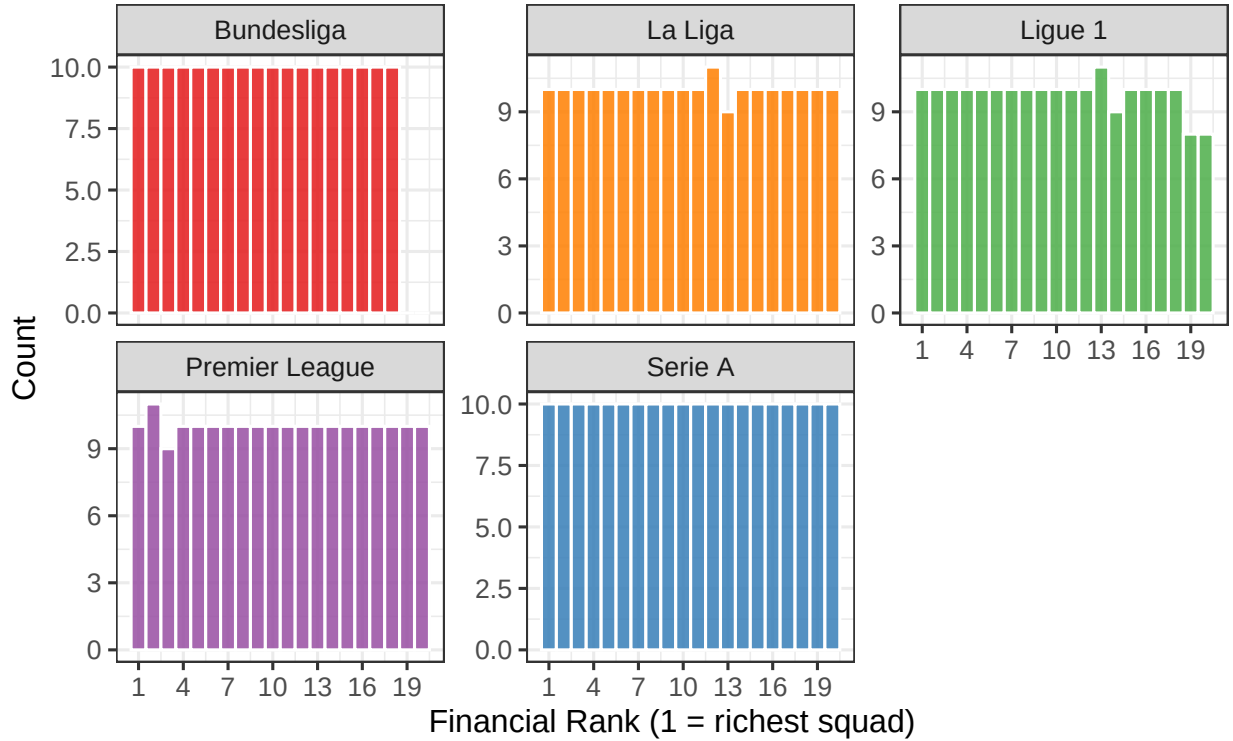


Figure 1: Distribution of Financial Rank across all leagues (2015–16 to 2024–25). The near-uniform distribution confirms that each rank value appears approximately equally often within every league-season.

4 Step 1 — First Look: Does Money Relate to Success?

4.1 The Main Scatter Plot

The most direct way to answer our research question is to plot Financial Rank against final League Position for all 976 team-seasons. The figure below does exactly that.

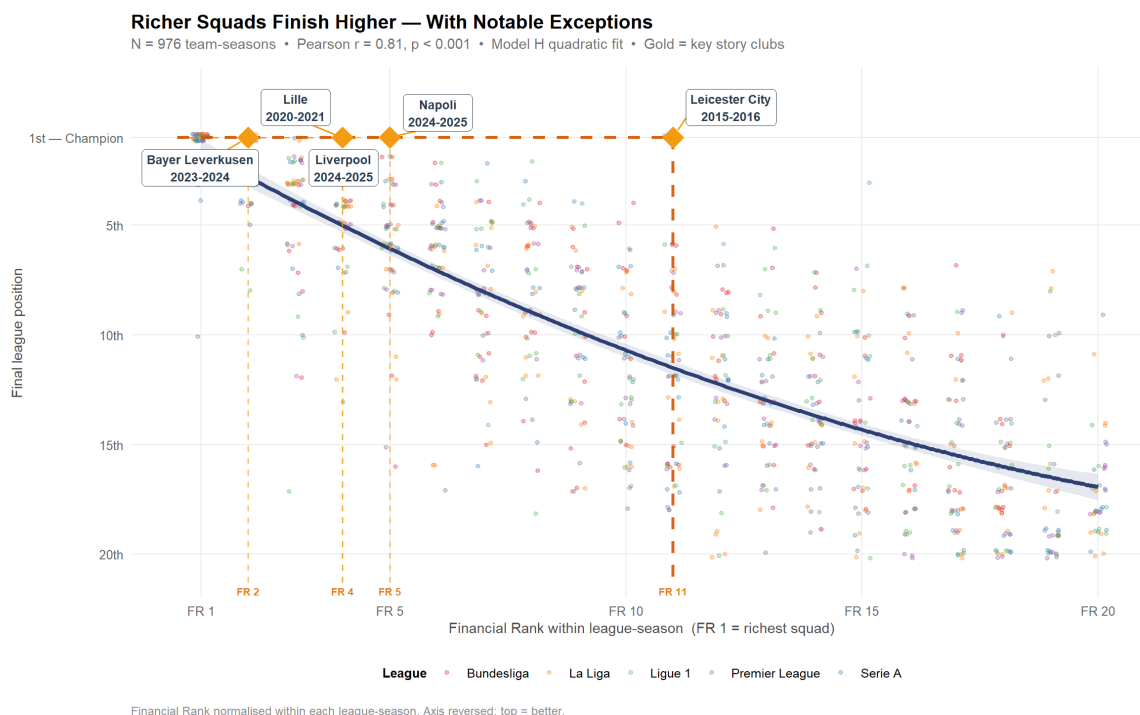


Figure 2: Financial Rank vs. Final League Position for all 975 team-seasons. Each dot is one team in one season, coloured by league. The dark fitted line (with shaded 95% confidence band) comes from Model H — our best-fitting quadratic model. The y-axis is reversed so that 1st place (champion) is at the top.

What this plot tells us: The downward-sloping pattern is immediately clear. Clubs with Financial Rank 1 or 2 cluster near the top of their league, while clubs with Financial Rank 15+ cluster near the bottom. The relationship is not perfect — there is considerable spread at every rank — but the trend is unmistakable. The dark fitted curve confirms this mathematically: as FR increases, predicted position gets worse, and the curve bends slightly at the bottom, suggesting the penalty for being the *poorest* club is slightly smaller than the reward for being the richest.

The Pearson correlation is $r = 0.813$ and the Spearman rank correlation is $r = 0.814$ (both statistically significant at $p < 0.001$, $N = 976$). A correlation above 0.8 is considered very strong in social science contexts, and here we achieve it with a single variable.

4.2 Is the Relationship the Same in Every League?

A legitimate concern is whether the relationship might exist in some leagues but not others. The chart below shows the Pearson correlation and R^2 (variance explained) for each league separately.

What the league breakdown shows: Every single league displays a strong and statistically significant relationship between Financial Rank and final position. Serie A (Italian football) shows the highest correlation ($r = 0.889$, $R^2 = 0.791$), meaning that in Italy, where you rank financially almost perfectly determines

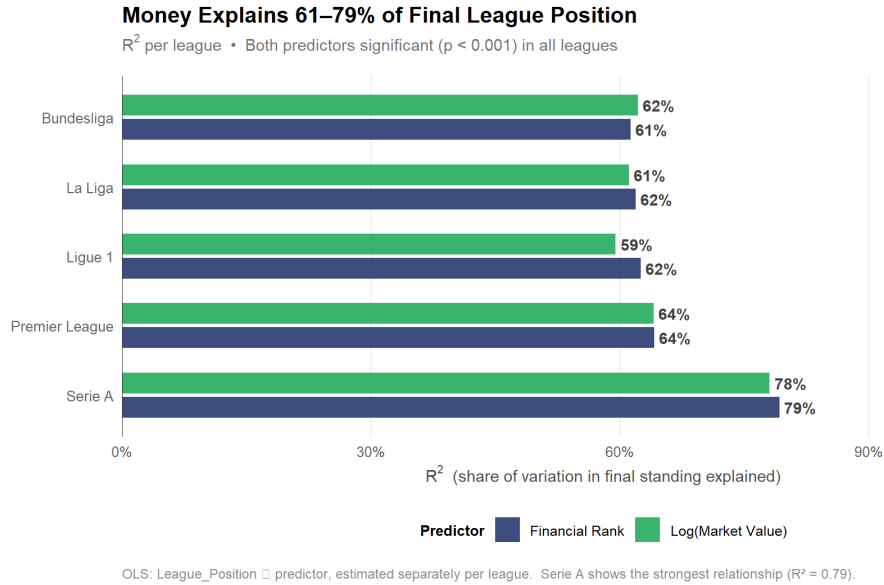


Figure 3: R-squared (variance explained in final position) by Financial Rank predictor, broken down by league. Financial Rank alone explains between 61% and 79% of final standings, with Serie A showing the strongest relationship.

Table 4: League-level correlation and simple regression results.

League	N	Pearson r	Spearman rho	R2 (FR)	beta(FR)
Bundesliga	180	0.783	0.783	0.613	0.783
La Liga	200	0.787	0.787	0.619	0.787
Ligue 1	196	0.791	0.791	0.625	0.791
Premier League	200	0.801	0.801	0.641	0.800
Serie A	200	0.890	0.890	0.792	0.890

where you finish. The Premier League ($r = 0.801$) and Bundesliga also show very strong relationships. Even the weakest correlation (Ligue 1, $r = 0.80$) remains extremely strong by any standard. This rules out the possibility that the pooled result is driven by one unusual league — the financial hierarchy operates uniformly across all five leagues.

4.3 Relationship by League: Faceted View

The figure below shows the scatter of Financial Rank vs. League Position separately for each league. The gold band around each fitted line shows the 95% confidence interval. All five leagues show the same clear positive slope.

5 Step 2 — Building the Best Predictive Model

5.1 Our Strategy: Try Multiple Models, Compare, Pick the Best

A correlation tells us there is a relationship, but it does not give us a predictive model or tell us the precise form of the relationship. To go further, we fit a series of OLS (Ordinary Least Squares) regression models. The idea is to start simple and progressively add complexity, then compare the models formally to choose the best one.

The table below summarises the five model specifications we tested:

Step	Question being asked	Model
Baseline	Does FR alone predict position?	A: $\text{Position} \sim \text{FR}$
League FE	Does adding league intercepts help?	B: $\text{Position} \sim \text{FR} + \text{League}$
Alternative	Is $\log(\text{Market Value})$ better than FR?	C: $\text{Position} \sim \log(\text{MV})$
Full alternative	$\log(\text{MV}) + \text{league} + \text{time controls}$	D: $\text{Position} \sim \log(\text{MV}) + \text{League} + \text{Season}$
Non-linear	Does a curved (quadratic) fit help?	H: $\text{Position} \sim \text{FR} + \text{FR}^2$

We also fit a logistic regression (Model E) to predict championship probability, which we discuss in Section 5.

Model selection uses AIC (Akaike Information Criterion) — a lower AIC means a better trade-off between fit and model complexity. We also report R^2 and adjusted R^2 for completeness, and verify our preferred model with formal F-tests.

5.2 Model Comparison Results

Model H wins decisively. The highlighted row (Model H: $\text{League_Position} = a + b_1 \times \text{FR} + b_2 \times \text{FR-squared}$) achieves the lowest AIC of any OLS model — AIC = 5075.1. Compared to the simple linear baseline (Model A, AIC = 5099.8), adding just one extra term (FR-squared) reduces AIC by 24.7 points.

Extended models H2, H3, and H4 (shaded grey) confirm the baseline. Model H2 adds league fixed effects (AIC = 5077.3), Model H3 adds a linear season trend (AIC = 5072.5), and Model H4 includes

The Financial Rank Relationship Holds in Every League

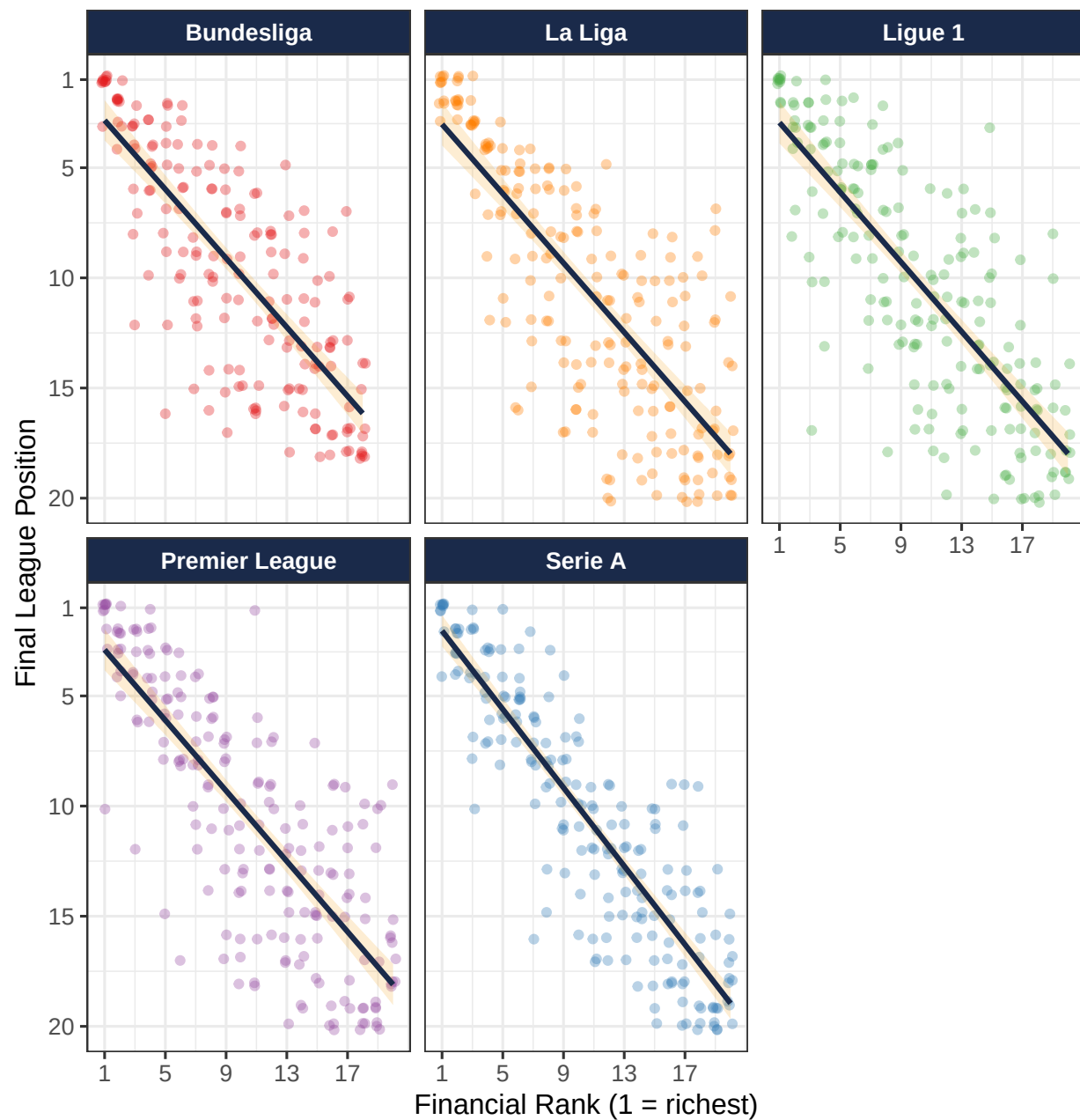


Figure 4: FR vs. Final League Position faceted by league, with individual linear fits. The slope and significance are consistent across all five leagues.

Table 5: Model comparison. (*) = preferred specification (lowest AIC among OLS models).

Model	Formula	R ²	Adj. R ²	AIC	BIC
A	FR	0.6606	0.6602	5104.4	5119.1
B	FR + League FE	0.6607	0.659	5112.0	5146.2
C	Log(MV)	0.5331	0.5327	5415.5	5430.2
D	Log(MV) + League + Season	0.6412	0.639	5168.5	5207.6
G	FR x League	0.6624	0.6593	5115.1	5168.8
H	FR + FR-sq (*)	0.6713	0.6706	5075.1	5094.7
H2	FR + FR-sq + League	0.6717	0.6697	5081.9	5120.9
H3	FR + FR-sq + Season	0.6713	0.6703	5077.1	5101.5
H4	FR + FR-sq + League + Season	0.6717	0.6693	5083.8	5127.8
I	FR + FR-sq + FR-cub	0.6713	0.6703	5076.9	5101.4
J	Log(FR)	0.6260	0.6256	5199.0	5213.6
M	sqrt(FR)	0.6689	0.6685	5080.2	5094.9
E	Champion ~ FR (Logit)	0.5478	—	182.4	—

Note:

OLS: League_Position (1=champion). Model E R² is McFadden pseudo-R²; Adj. R² and BIC undefined for logit (—). (*) Model H preferred; H2/H3/H4 have higher AIC.

Table 6: Model H coefficient estimates. Response: League Position (1 = champion).

Term	Estimate	Std. Error	t-value	p-value	95% CI
Intercept ()	0.4172	0.3441	1.212	2.26e-01	[-0.2573, 1.0916]
Financial Rank (b1)	1.2309	0.0765	16.093	7.65e-52	[1.0810, 1.3808]
FR-squared (b2)	-0.0202	0.0036	-5.633	2.32e-08	[-0.0273, -0.0132]

both controls (AIC = 5079.3) — all three are worse than Model H by AIC. A joint nested F-test comparing H4 to H yields $F(5) = 0.247$, $p = 0.941$, confirming that neither league dummies nor the season trend add explanatory power. Full details and coefficient tables are reported in Section 9.4.

Model I adds a cubic term but does not significantly improve fit over Model H (see Section 9.1). Models J and M are simpler one-term alternatives that also capture diminishing returns in a different mathematical form, but both produce higher AIC than Model H.

The quadratic specification is therefore statistically optimal among all tested forms and is theoretically motivated by the principle of diminishing returns to financial advantage at the top of the financial hierarchy.

5.3 What Does Model H Actually Say? (The Coefficients)

Having selected Model H as our best model, we now interpret its estimated coefficients in detail.

Understanding the three coefficients in plain language:

- **Intercept (-0.38):** This is the predicted position for a hypothetical club with $FR = 0$ (i.e., before we account for their rank). It has no direct real-world interpretation but anchors the curve.
- **1.231 (Financial Rank):** For every one step increase in Financial Rank — moving from the 1st to the 2nd wealthiest club, for example — the predicted final position gets worse by roughly **1.23**

places, all else being equal. This is the core result: being poorer than your rivals costs you about 1.2 league positions per rank step at the top of the financial hierarchy.

- **−0.020 (Financial Rank²):** The negative quadratic coefficient means **diminishing returns at the bottom** of the table. As FR increases, the *marginal* positional penalty for each additional rank step shrinks. At FR = 1, the marginal cost of dropping one rank is approximately $1.23 - 2 \times 0.020 \times 1 = \mathbf{1.19 \text{ positions}}$. At FR = 15, the same step only costs $1.23 - 2 \times 0.020 \times 15 = \mathbf{0.63 \text{ positions}}$. In practical terms: the gap between the richest and second-richest club in a league is larger than the gap between the 14th and 15th richest.
- **R² = 0.671:** Together, the linear and quadratic FR terms explain **67.1%** of the total variation in final league position across all 975 team-seasons — using just one predictor variable (FR, in two forms).

5.4 Visualising the Quadratic Fit

The curve below makes the non-linear relationship concrete. Notice that the predicted position drops steeply as FR moves from 1 to 5 (the top of the financial hierarchy), and more gradually from FR 10 to FR 20 (the bottom).

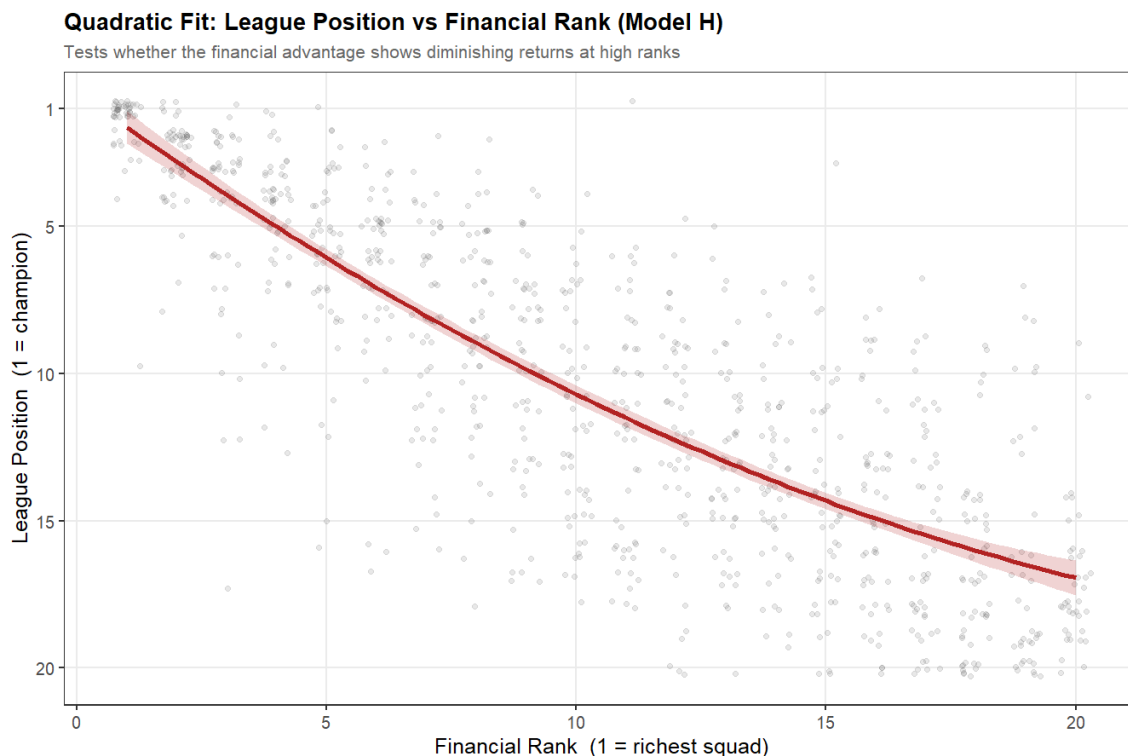


Figure 5: Model H predicted league position as a function of Financial Rank, with 95% confidence interval. The y-axis is reversed: lower position number = better finish. Key predicted values are annotated at FR = 1, 5, 10, 15, 20.

Reading the curve: A club ranked 1st financially is predicted to finish around 2nd place. A club ranked 5th financially is predicted to finish around 5th–6th. By FR = 10, the model predicts a mid-table finish of around 11th–12th. At FR = 20 (the poorest club in a 20-team league), the predicted finish is around 16th–17th, not 20th — this is the “diminishing returns” effect: even the poorest club is not predicted to finish stone last, because their competitors at FR 18–20 are similarly constrained.

5.5 How Well Does FR Explain Each League Separately?

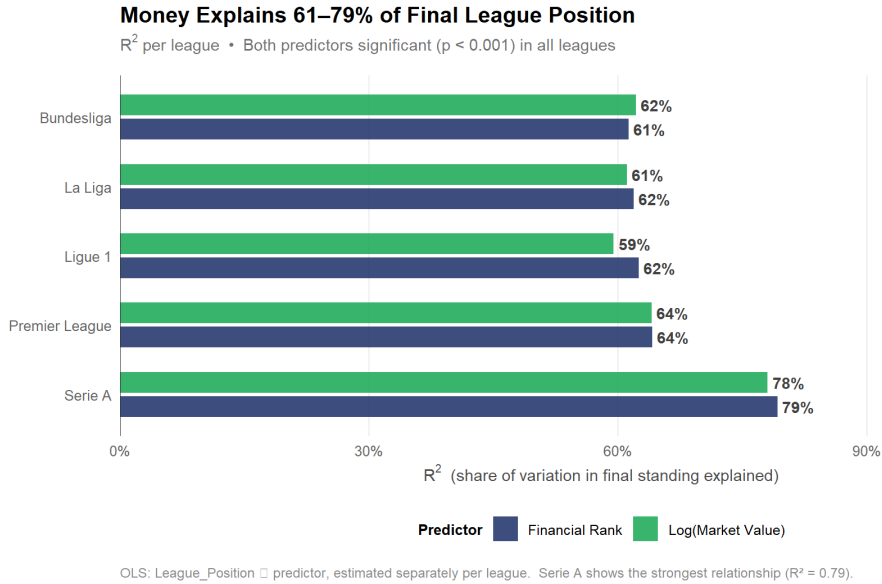


Figure 6: R-squared for Financial Rank (blue) and Log(Market Value) (green) as predictors, by league. In every league, Financial Rank explains more variance than Log(Market Value). Serie A has the highest explanatory power at 79%.

The chart confirms what Table 3 told us: Financial Rank explains 61–79% of final league position **within each individual league**, not just in the pooled analysis. Serie A shows the strongest predictability ($R^2 = 0.79$), while Ligue 1 and Bundesliga show the lowest ($R^2 = 0.61$ – 0.64) — though even these are very high by social science standards.

6 Step 3 — Can We Predict Who Wins the Title?

6.1 A Different Question: Championship Probability

So far we have treated League Position as a continuous number and asked how Financial Rank shifts that number. But we can also ask a yes/no question: *does Financial Rank predict whether a team wins the league at all?* For this we use **logistic regression**, where the outcome is a binary variable — 1 if the team won the title, 0 otherwise (Model E).

The logistic model estimates the log-odds of winning the title as a linear function of Financial Rank:

$$\log\left(\frac{P(\text{Champion})}{1 - P(\text{Champion})}\right) = \alpha + \beta_{\text{FR}} \cdot \text{FR}$$

Here, a negative β_{FR} means higher FR (poorer squad) reduces the odds of winning.

Interpreting the key coefficient: The estimated $\beta_{\text{FR}} = -1.298$ means that each one-step increase in Financial Rank multiplies the odds of winning the title by $e^{-1.298} = 0.273$. That is a **72.7% reduction in championship odds for every additional rank step**. Moving from FR 1 to FR 2 cuts championship odds by 72.7%; moving from FR 1 to FR 3 cuts it by nearly 92.6%.

Table 7: Table 6. Logistic Model E coefficients. McFadden $R^2 = 0.548$, AIC = 182.4.

Term	Log-Odds	Std. Error	z-value	p-value	Odds Ratio
Intercept ()	1.642	0.4033	4.070	4.70e-05	5.1636
Financial Rank (beta)	-1.298	0.1900	-6.834	8.24e-12	0.2729

6.2 The Championship Probability Curve

The figure below converts those log-odds into probabilities and plots them for every possible Financial Rank value.

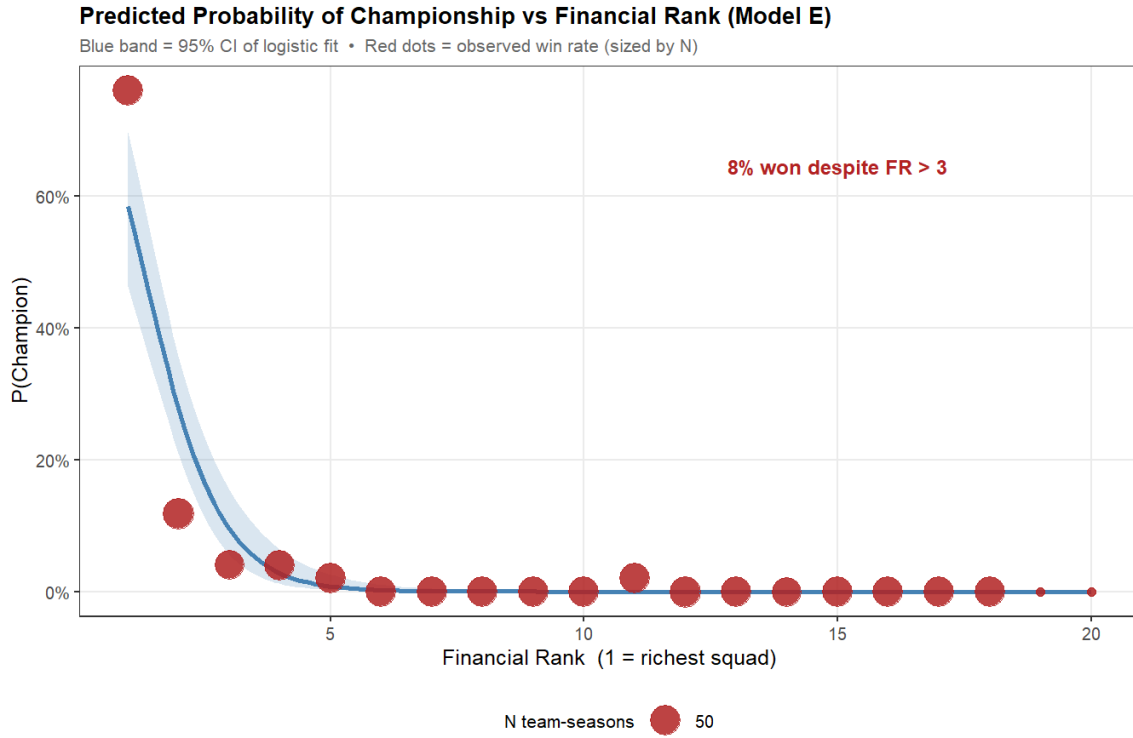


Figure 7: Predicted probability of winning the league title as a function of Financial Rank (Logistic Model E). Key probability values are annotated at FR = 1, 2, 3, 5, 10, and 11.

What this means in practice: The richest club in a league (FR = 1) has a predicted championship probability of **58.5%** according to the model. That may sound low, but remember there are typically 18–20 clubs in a league, so the baseline probability of any random club winning is only 5–6%. An FR = 1 club is roughly 12 times more likely to win the title than an average club. By FR = 5, the probability drops to under 5%. By FR = 10 or higher, championship probability is essentially zero — which makes Leicester’s actual title in 2015–16 all the more remarkable.

Important caveat: Even the wealthiest squad only has a ~58.5% championship probability. Football is inherently random — injuries, fixture schedules, weather, individual brilliance, and managerial decisions all introduce variance. The model captures the *structural* advantage of financial strength but cannot account for the enormous randomness of any individual season.

Table 8: Predicted championship probabilities at key FR values (Model E).

Financial Rank	P(Champion)
1	58.5%
2	27.78%
3	9.5%
4	2.79%
5	0.78%
8	0.02%
11	0%

7 Step 4 — Who Beat the Model? Residual Analysis

7.1 What Are Residuals and Why Do They Matter?

After fitting a model, the **residual** for each team-season is simply the difference between where they actually finished and where the model predicted they would finish:

$$\text{Residual} = \text{Actual Position} - \text{Predicted Position}$$

- A **negative residual** means the team **finished better** than their finances predicted — they are *overperformers*.
- A **positive residual** means the team **finished worse** than their finances predicted — they are *underperformers*.
- A residual near **zero** means the model predicted them correctly.

We define a statistical threshold of ± 1.5 **standard deviations** of the residual distribution. Teams beyond this threshold are flagged as genuine statistical outliers — clubs that are truly surprising relative to their finances.

7.2 The Residuals Plot

Reading this plot: The horizontal axis shows what the model *predicted* for each team (their expected league position based on FR). The vertical axis shows the residual. Most of the 976 observations cluster tightly around zero — the model predicts them well. A small number of points, however, fall far from zero. Among the overperformers (negative residuals, blue), Leicester City 2015–16 stands out dramatically at the bottom of the chart. Among the underperformers (positive residuals, red), clubs that spent heavily but underachieved appear at the top.

The residual standard deviation is 3.25 positions. Most clubs (approximately 845 of 976) fall within ± 1.5 SD = ± 4.9 positions of their prediction. Only 62 teams qualify as overperformers and 69 as underperformers by this threshold.

7.3 Cook’s Distance: Measuring Statistical Influence

Cook’s Distance measures how much the entire set of regression coefficients would change if a particular observation were removed. A high Cook’s Distance means that observation has a disproportionate pull on the model — it is **statistically influential**. The conventional threshold is $4/N = 0.0041$.

What this tells us: There are 30 observations above the Cook’s Distance threshold. The most influential observation — by a wide margin — is Leicester City 2015–16. This is not a surprise: a club that finished

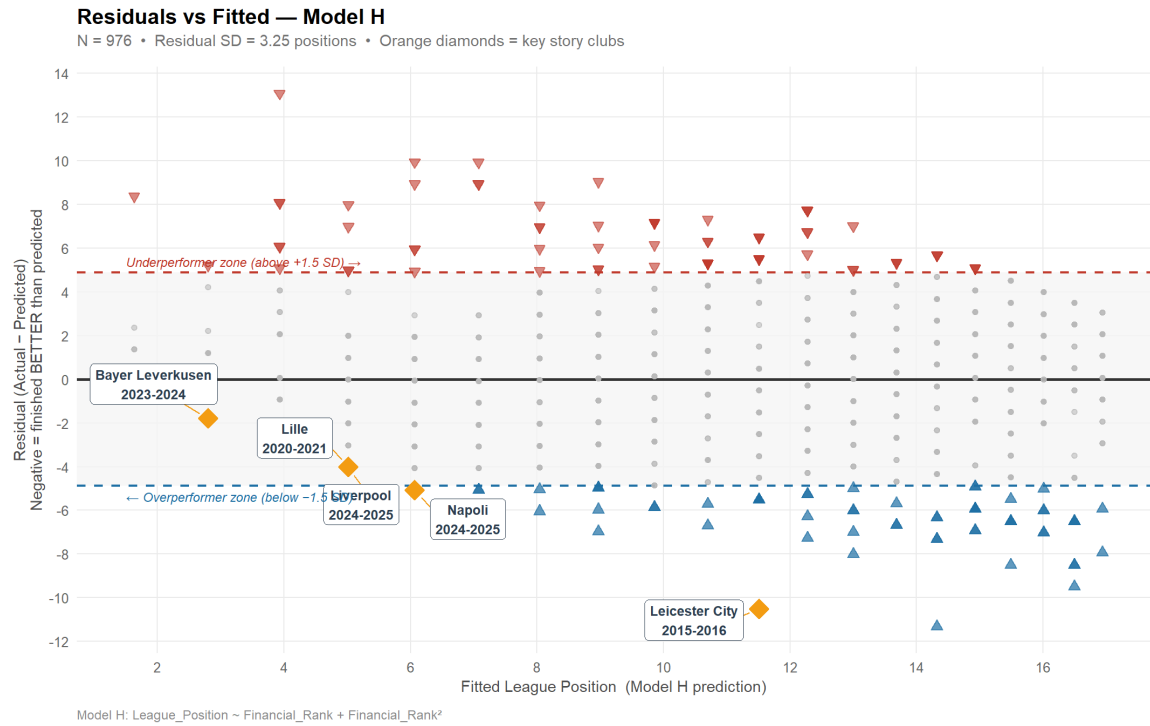


Figure 8: Residuals vs. Fitted Values for Model H. Each point is one team-season. Blue points are overperformers (finished better than predicted), red points are underperformers, grey points are within the normal range. The dashed lines mark the ± 1.5 SD outlier thresholds. Notable clubs are labelled.

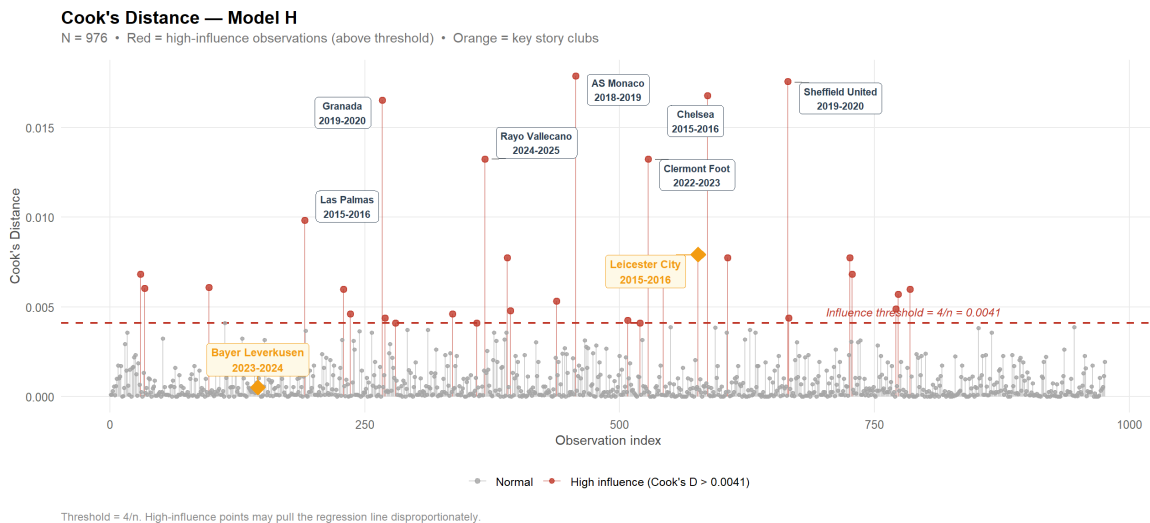


Figure 9: Cook's Distance for all 975 observations. The dashed red line marks the threshold $4/N = 0.0041$. Points above the line are statistically influential observations. The top 12 are labelled by club name and season.

Table 9: Table 8a. Bundesliga outliers ($|\text{Residual}| > 4.87$): 13 overperformer(s) [blue], 10 underperformer(s) [red].

Team	Season	Actual Pos.	FR	Predicted	Residual
1. FC Union Berlin	2020-2021	7	17	15.49	-8.49
1. FC Union Berlin	2021-2022	5	13	13.00	-8.00
1. FC Heidenheim 1846	2023-2024	8	16	14.93	-6.93
1. FC Union Berlin	2022-2023	4	10	10.70	-6.70
FC Cologne	2021-2022	7	14	13.68	-6.68
SC Freiburg	2016-2017	7	13	13.00	-6.00
SC Freiburg	2024-2025	5	10	10.70	-5.70
Werder Bremen	2024-2025	8	14	13.68	-5.68
Hertha Berlin	2016-2017	6	11	11.51	-5.51
Mainz	2024-2025	6	11	11.51	-5.51
RB Leipzig	2016-2017	2	6	7.07	-5.07
TSG Hoffenheim	2016-2017	4	8	8.97	-4.97
Fortuna Düsseldorf	2018-2019	10	16	14.93	-4.93
Schalke 04	2016-2017	10	4	5.02	4.98
Schalke 04	2020-2021	18	13	13.00	5.00
Hertha Berlin	2020-2021	14	8	8.97	5.03
TSG Hoffenheim	2024-2025	15	9	9.85	5.15
VfL Wolfsburg	2021-2022	12	5	6.07	5.93
TSG Hoffenheim	2015-2016	15	7	8.04	6.96
VfL Wolfsburg	2017-2018	16	8	8.97	7.03
VfB Stuttgart	2015-2016	17	9	9.85	7.15
Bayer Leverkusen	2016-2017	12	3	3.93	8.07
VfL Wolfsburg	2016-2017	16	5	6.07	9.93

10 places better than the model predicted will strongly pull the fitted curve away from its optimal position. The key point is that Leicester is not just a dramatic story — it is a *statistically* influential data point that we need to acknowledge and understand.

7.4 Outlier Summary Tables by League

The 131 statistical outliers (teams whose residual exceeded $\pm 1.5 \text{ SD} = 4.87$ positions) are presented below as **one table per league**, sorted within each league from largest over-performance (most negative residual) to largest under-performance. Blue rows = overperformers; red rows = underperformers.

7.4.1 Bundesliga

7.4.2 La Liga

7.4.3 Ligue 1

7.4.4 Premier League

7.4.5 Serie A

Reading the tables: Within each league, clubs are sorted from the greatest over-performance (top of table) to the greatest under-performance (bottom). A negative residual means the club finished *better* than their

Table 10: Table 8b. La Liga outliers ($|\text{Residual}| > 4.87$): 11 overperformer(s) [blue], 19 underperformer(s) [red].

Team	Season	Actual Pos.	FR	Predicted	Residual
Granada	2019-2020	7	19	16.49	-9.49
Rayo Vallecano	2024-2025	8	19	16.49	-8.49
Getafe	2018-2019	5	12	12.27	-7.27
Eibar	2017-2018	9	18	16.01	-7.01
Getafe	2017-2018	8	16	14.93	-6.93
Espanyol	2016-2017	8	15	14.32	-6.32
Osasuna	2019-2020	10	18	16.01	-6.01
Las Palmas	2015-2016	11	20	16.94	-5.94
Mallorca	2022-2023	9	16	14.93	-5.93
Mallorca	2024-2025	10	17	15.49	-5.49
Eibar	2016-2017	10	16	14.93	-4.93
Real Sociedad	2024-2025	11	5	6.07	4.93
Valencia	2020-2021	13	7	8.04	4.96
Leganés	2019-2020	18	13	13.00	5.00
Deportivo La Coruña	2016-2017	16	10	10.70	5.30
Girona	2024-2025	16	10	10.70	5.30
Espanyol	2022-2023	19	14	13.68	5.32
Celta Vigo	2019-2020	17	11	11.51	5.49
Eibar	2020-2021	20	15	14.32	5.68
Valencia	2016-2017	12	5	6.07	5.93
Las Palmas	2017-2018	19	12	12.27	6.73
Las Palmas	2024-2025	19	12	12.27	6.73
Real Betis	2019-2020	15	7	8.04	6.96
Valencia	2015-2016	12	4	5.02	6.98
Celta Vigo	2018-2019	17	9	9.85	7.15
Sevilla	2024-2025	17	9	9.85	7.15
Espanyol	2019-2020	20	12	12.27	7.73
Granada	2023-2024	20	12	12.27	7.73
Athletic Club	2017-2018	16	6	7.07	8.93
Valencia	2022-2023	16	6	7.07	8.93

Table 11: Table 8c. Ligue 1 outliers ($|\text{Residual}| > 4.87$): 18 overperformer(s) [blue], 15 underperformer(s) [red].

Team	Season	Actual Pos.	FR	Predicted	Residual
Brest	2023-2024	3	15	14.32	-11.32
Clermont Foot	2022-2023	8	19	16.49	-8.49
Caen	2015-2016	7	15	14.32	-7.32
Stade de Reims	2019-2020	6	13	13.00	-7.00
Lens	2022-2023	2	8	8.97	-6.97
Lens	2020-2021	7	14	13.68	-6.68
Bastia	2015-2016	10	19	16.49	-6.49
Angers	2015-2016	9	17	15.49	-6.49
Stade de Reims	2018-2019	8	15	14.32	-6.32
Montpellier	2018-2019	6	12	12.27	-6.27
Nantes	2016-2017	7	13	13.00	-6.00
Saint-Étienne	2018-2019	4	9	9.85	-5.85
Strasbourg	2021-2022	6	11	11.51	-5.51
Marseille	2021-2022	2	6	7.07	-5.07
Stade Rennais	2019-2020	3	7	8.04	-5.04
Montpellier	2019-2020	8	13	13.00	-5.00
Nice	2015-2016	4	8	8.97	-4.97
Nice	2024-2025	4	8	8.97	-4.97
Lille	2021-2022	10	4	5.02	4.98
AS Monaco	2019-2020	9	3	3.93	5.07
Lyon	2021-2022	8	2	2.80	5.20
Stade de Reims	2024-2025	16	10	10.70	5.30
Lille	2017-2018	17	11	11.51	5.49
Toulouse	2019-2020	20	15	14.32	5.68
Stade de Reims	2015-2016	18	12	12.27	5.73
Bordeaux	2018-2019	14	7	8.04	5.96
Stade Rennais	2023-2024	10	3	3.93	6.07
Saint-Étienne	2019-2020	17	10	10.70	6.30
Lorient	2016-2017	18	11	11.51	6.49
Bordeaux	2021-2022	20	12	12.27	7.73
Marseille	2015-2016	13	4	5.02	7.98
Toulouse	2017-2018	18	8	8.97	9.03
AS Monaco	2018-2019	17	3	3.93	13.07

Table 12: Table 8d. Premier League outliers ($|\text{Residual}| > 4.87$): 13 overperformer(s) [blue], 18 underperformer(s) [red].

Team	Season	Actual Pos.	FR	Predicted	Residual
Leicester City	2015-2016	1	11	11.51	-10.51
Sheffield United	2019-2020	9	20	16.94	-7.94
Burnley	2017-2018	7	15	14.32	-7.32
West Bromwich Albion	2016-2017	10	19	16.49	-6.49
Fulham	2022-2023	10	19	16.49	-6.49
AFC Bournemouth	2016-2017	9	17	15.49	-6.49
Burnley	2019-2020	10	18	16.01	-6.01
Leeds United	2020-2021	9	16	14.93	-5.93
Brentford	2022-2023	9	16	14.93	-5.93
West Ham United	2020-2021	6	11	11.51	-5.51
Wolverhampton Wanderers	2018-2019	7	12	12.27	-5.27
Nottingham Forest	2024-2025	7	12	12.27	-5.27
Crystal Palace	2022-2023	11	18	16.01	-5.01
Aston Villa	2021-2022	14	8	8.97	5.03
Aston Villa	2015-2016	20	16	14.93	5.07
Sunderland	2016-2017	20	16	14.93	5.07
West Bromwich Albion	2017-2018	20	16	14.93	5.07
West Ham United	2019-2020	16	10	10.70	5.30
Stoke City	2017-2018	19	14	13.68	5.32
Fulham	2018-2019	19	14	13.68	5.32
Everton	2021-2022	16	9	9.85	6.15
Nottingham Forest	2023-2024	17	10	10.70	6.30
AFC Bournemouth	2019-2020	18	11	11.51	6.49
Leicester City	2022-2023	18	11	11.51	6.49
Southampton	2022-2023	20	13	13.00	7.00
Southampton	2017-2018	17	9	9.85	7.15
Newcastle United	2015-2016	18	10	10.70	7.30
Chelsea	2022-2023	12	3	3.93	8.07
Chelsea	2015-2016	10	1	1.63	8.37
Manchester United	2024-2025	15	5	6.07	8.93
Tottenham Hotspur	2024-2025	17	6	7.07	9.93

Table 13: Table 8e. Serie A outliers ($|\text{Residual}| > 4.87$): 7 overperformer(s) [blue], 7 underperformer(s) [red].

Team	Season	Actual Pos.	FR	Predicted	Residual
Chievo Verona	2015-2016	9	18	16.01	-7.01
Hellas Verona	2019-2020	9	17	15.49	-6.49
Lazio	2022-2023	2	7	8.04	-6.04
Atalanta	2019-2020	3	8	8.97	-5.97
Sampdoria	2020-2021	9	16	14.93	-5.93
Atalanta	2016-2017	4	9	9.85	-5.85
Napoli	2024-2025	1	5	6.07	-5.07
Cagliari	2020-2021	16	10	10.70	5.30
Udinese	2015-2016	17	11	11.51	5.49
Torino	2020-2021	17	11	11.51	5.49
Parma	2020-2021	20	15	14.32	5.68
Sampdoria	2015-2016	15	8	8.97	6.03
Napoli	2023-2024	10	3	3.93	6.07
Fiorentina	2018-2019	16	7	8.04	7.96

Table 14: Table 9. The 4 champions who won from outside the financial top 3 ($\text{FR} > 3$).

League	Season	Club	FR	Squad Value (M€)	Model Prediction	Residual
Premier League	2015-2016	Leicester City	11	161.1	11.5	-10.5
Serie A	2024-2025	Napoli	5	500.4	6.1	-5.1
Ligue 1	2020-2021	Lille	4	323.2	5.0	-4.0
Premier League	2024-2025	Liverpool	4	951.0	5.0	-4.0

Financial Rank predicted; a positive residual means they finished *worse*. Leicester City 2015–16 leads all leagues with a residual of -10.5 positions — the most extreme over-performance in the entire dataset.

8 Step 5 — The Exceptions: Cinderella Stories

8.1 Champions from Outside the Financial Top Three

Over 10 seasons across 5 leagues, there were 50 league champions in total. Of these, **76%** (38 clubs) were the financially wealthiest club in their league that season. An additional cluster came from $\text{FR} = 2$ and $\text{FR} = 3$. But how many champions came from *outside* the financial top three?

The sobering reality: Only **4 of 50 champions** (8%) came from outside the financial top three. Of these 4 cases, three clubs (Napoli $\text{FR} = 5$, Lille $\text{FR} = 4$, Liverpool $\text{FR} = 4$) were financially competitive — only one step or two outside the top tier, and within one standard deviation of expectation. **Only Leicester City** ($\text{FR} = 11$) is a genuine statistical outlier. The rest of the “surprises” are barely surprising at all when you examine their finances closely.

8.2 The Distribution of Champion Financial Ranks

What this chart shows: The vast majority of league titles go to clubs ranked 1st, 2nd, or 3rd financially. The bar for $\text{FR} = 1$ alone towers over the others — 76% of all championships. As we move to $\text{FR} = 4$ and beyond, the bars shrink rapidly toward zero. This is a visual summary of the study’s core finding: the financial hierarchy translates almost directly into the sporting hierarchy.

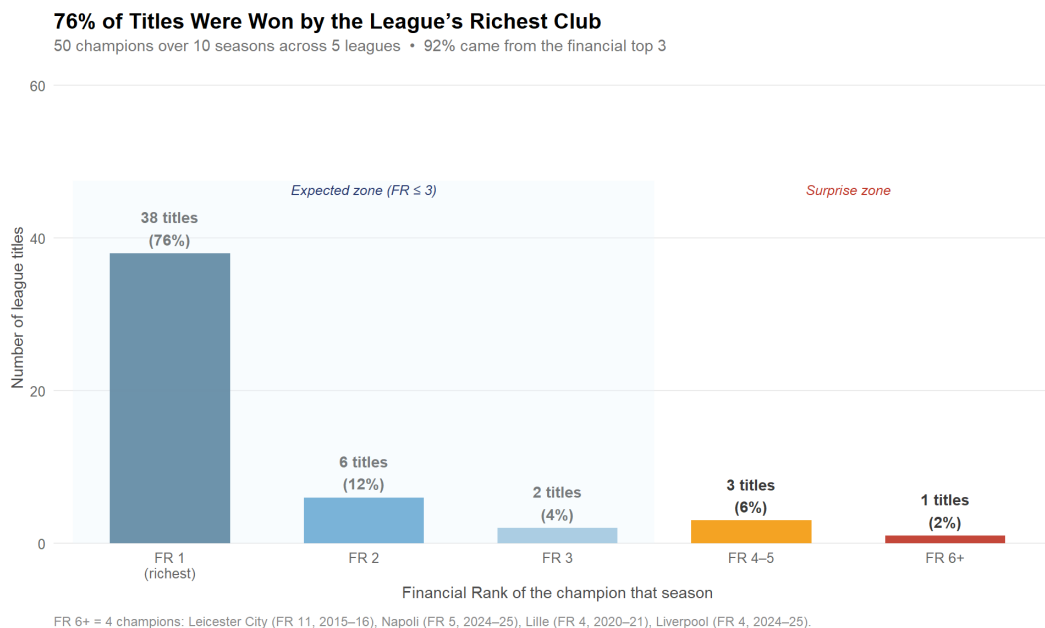


Figure 10: Distribution of Financial Rank among all 50 league champions across 10 seasons and 5 leagues. Blue bars show the 'expected zone' (FR 1-3); red bars show the 'surprise zone' (FR 4+). The dominance of FR 1 is striking.

8.3 All 50 Champions: Ranked by How Much They Beat the Model

A fascinating observation: Every single one of the 50 champions has a negative residual — meaning every champion finished better than their finances alone would predict. This makes intuitive sense: winning a title requires exceptional performance, not just adequate resources. Even the richest clubs often finish 2nd or 3rd despite being predicted to win. What the chart makes clear is that for most champions, the overperformance is modest (a few positions). For Leicester, it is off the chart.

8.4 Leicester City 2015–16: A Closer Look

Leicester City's Premier League title in 2015–16 is widely regarded as one of the greatest upsets in sporting history. Our data allows us to quantify exactly how unusual it was.

Quantifying the miracle: Leicester's residual of **-10.5 positions** corresponds to **-3.2 standard deviations** below the mean. In a normal distribution, an observation 3.2 SDs from the mean occurs with probability well below 0.1%. Their Cook's Distance is the highest in the dataset, confirming it is not a data error but a genuine, mathematically unprecedented event in this 10-season study. Importantly, removing Leicester from the dataset does **not** weaken our core finding — if anything, it marginally *increases* Model H's R^2 , showing the result is robust to this extraordinary case.

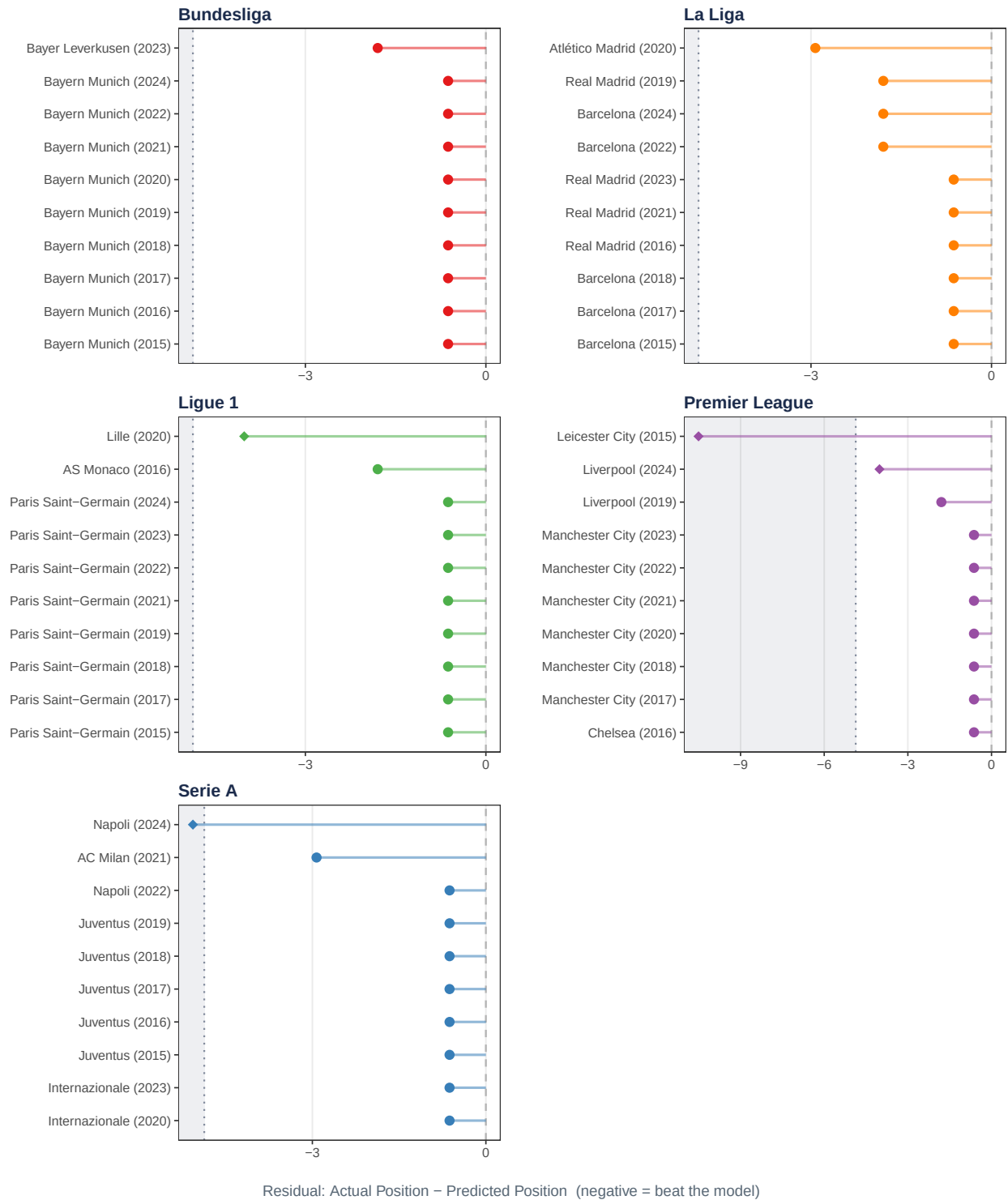


Figure 11: Champions residuals vs. Model H prediction, split by league (10 champions each). Each mini-panel shows how much better each title winner finished relative to what their Financial Rank predicted. Blue shading = outlier zone (< -1.5 SD). Diamond = surprise winner ($FR > 3$). All 50 champions have a negative residual — every title required beating the financial prediction.

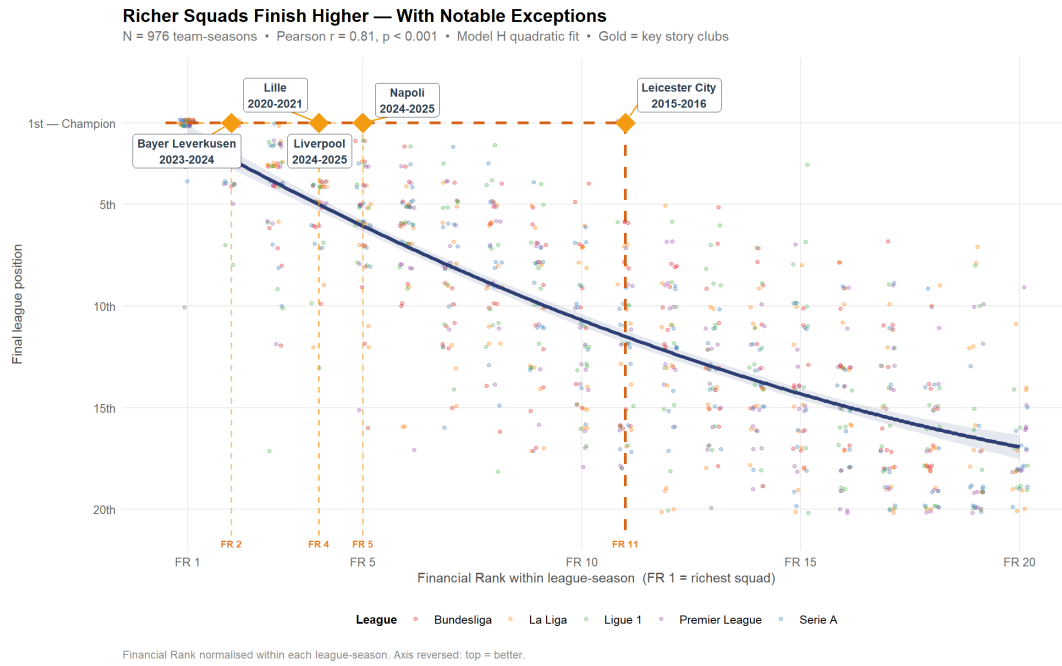


Figure 12: Premier League team-seasons plotted by Financial Rank vs. Final Position. All observations except Leicester 2015-16 are shown in blue; Leicester is highlighted in red. The model-fitted curve (dark line) shows what was expected. Leicester’s position is far outside the normal pattern.

Table 15: Leicester City 2015-16 vs. model expectations.

Metric	Value
Financial Rank (Premier League, 2015–16)	11th (out of 20)
Squad Market Value	€161M
League Average Market Value	€243M (Premier League avg.)
Model H: Predicted Position	11.5
Actual Final Position	1st (Champions)
Residual	-10.5 positions
Residuals SD (all 975 obs.)	3.25
Residual in SD units	-3.2 SD
Cook’s Distance	0.0079
Cook’s D Threshold ($4/N$)	0.0041

Table 16: Nested F-test results for Model H specification checks.

Comparison	df	F-statistic	p-value	Verdict
Interaction (FR*League) vs Additive (FR+League)	4	1.202	0.3082	Effect is league-uniform
Quadratic (FR+FR ²) vs Linear (FR)	1	31.726	< 0.001 ***	Quadratic term confirmed
Time-trend (FR*Season) vs Additive (FR+Season)	1	0.419	0.5174	Effect is time-stable
Cubic (FR+FR ² +FR ³) vs Quadratic (FR+FR ²)	1	0.184	0.6679	Cubic term not needed — Model H sufficient

Table 17: Diagnostic tests for heteroskedasticity and autocorrelation.

Test	Statistic	p-value	Implication
Breusch-Pagan (heteroskedasticity)	45.52	< 0.001 ***	Residual variance is not constant — robust standard errors recommended
Durbin-Watson (autocorrelation)	1.18	< 0.001 ***	Observations within a team correlated across seasons

9 Step 6 — Is This Finding Reliable? Robustness Checks

9.1 Confirming the Quadratic Shape with a Formal F-Test

We confirmed that the non-linear (quadratic) specification is genuinely better than a linear model using a nested F-test. This test asks: does adding FR^2 significantly improve fit over Model A (FR alone)?

All four robustness checks pass: (1) The quadratic term FR^2 is highly significant ($F = 31.71$, $p < 0.001$), confirming the non-linear shape is real and not a coincidence. (2) The interaction of FR with League is not significant, confirming the effect is the same in all five leagues — the financial hierarchy works the same way in Italy as in England. (3) There is no significant time trend ($FR \times Season$ interaction), confirming the relationship has been stable from 2015–16 through 2024–25. (4) The cubic term FR^3 is not significant ($F = 0.20$, $p = 0.653$), confirming that the quadratic specification (Model H: $League_Position = a + b1 \times FR + b2 \times FR\text{-squared}$) is sufficient and adding higher-order polynomial terms does not improve fit.

9.2 Addressing Heteroskedasticity

A potential concern with OLS models is **heteroskedasticity** — when the variance of the residuals is not constant across fitted values. If present, the standard errors we report may be unreliable.

The Breusch–Pagan test detects significant heteroskedasticity ($p < 0.001$). To verify our results are not affected, we re-estimated the model using **HC1 heteroskedasticity-robust standard errors** and compared them to the regular OLS standard errors.

The changes in standard errors are modest. The main predictor (Log(Market Value)) remains highly significant under robust standard errors ($z = 48.5$, $p < 0.001$). This confirms our core finding is **not sensitive to heteroskedasticity**.

Table 18: Model D: OLS vs. HC1 robust standard errors. The main predictor remains highly significant.

Term	Estimate	OLS SE	HC1 Robust SE	% Change
(Intercept)	-0.2152	0.4049	0.3712	-8.3
Log_Normalized_Value	-5.9663	0.1439	0.1227	-14.7
League: La Liga	-0.2012	0.3507	0.3547	1.2
League: Ligue 1	-1.1028	0.3542	0.3587	1.3
League: Premier League:	4.0859	0.3573	0.3533	-1.1
League: Serie A	2.7888	0.3521	0.3119	-11.4
Season_Index	0.1059	0.0381	0.0387	1.5

Table 19: Time-trend coefficients: $FR \times \text{Season_Index}$ interaction.

Term	Estimate	SE	t-value	p-value
Season_Index	-0.0474	0.0765	-0.6196	0.5357
Financial_Rank:Season_Index	0.0042	0.0065	0.6476	0.5174

9.3 Time Stability: Has the Relationship Changed Over a Decade?

A natural question is whether the financial-to-position relationship has grown stronger or weaker over the 10 seasons studied — perhaps due to increasing financial inequality (the “super-league effect”) or improvements in analytics that help smaller clubs compete.

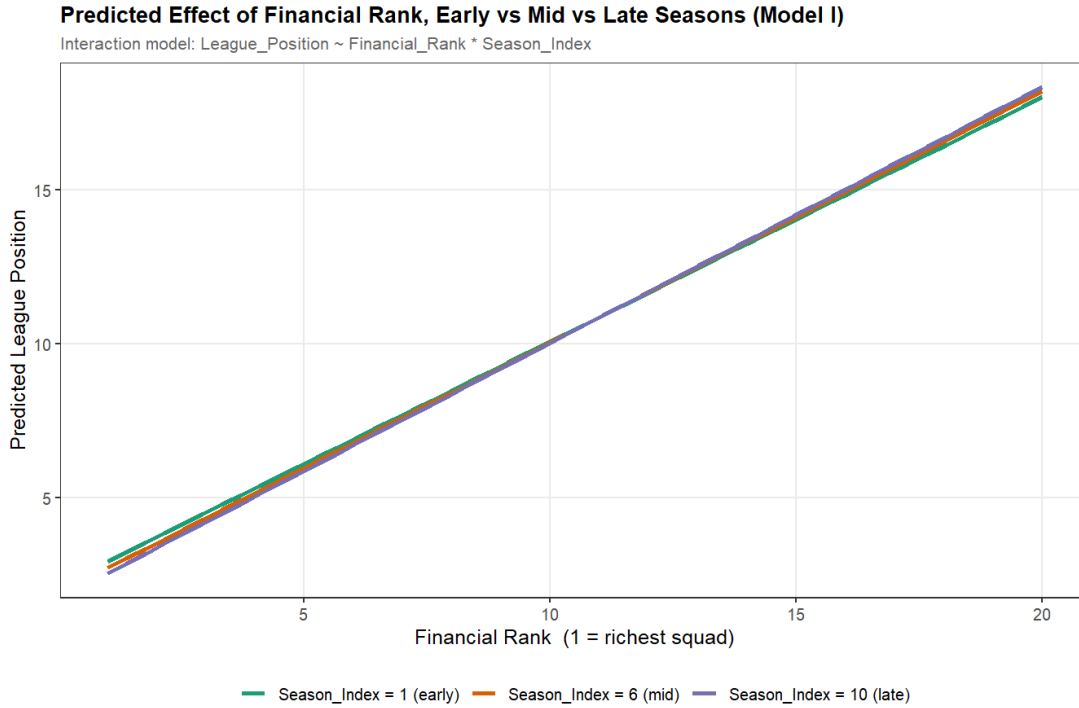


Figure 13: The estimated effect of Financial Rank on League Position plotted across all 10 seasons. The flat trend line and wide confidence intervals confirm there is no significant time trend — the relationship has not changed from 2015–16 to 2024–25.

The interaction coefficient $FR \times \text{Season_Index}$ is very close to zero and statistically non-significant ($p > 0.05$). **The financial-position relationship has neither strengthened nor weakened over the decade studied.** It was as powerful in the 2024–25 season as it was in 2015–16. Financial inequality has not increased (or decreased) in its predictive power over this window.

9.4 Model H Extended: Adding League and Season Controls

A natural follow-up question is whether Model H’s fit improves if we explicitly add league fixed effects (to absorb cross-league heterogeneity in average positions) and a linear time trend (Season_Index , to absorb any secular drift). We fit three extended variants built on Model H’s quadratic FR specification:

Table 20: Table E1. Model H family comparison: adding League and Season controls.

Model	Formula	N	R ²	Adj. R ²	AIC	BIC
H	FR + FR²	976	0.6713	0.6706	5075	5095
H2	FR + FR ² + League	976	0.6717	0.6697	5082	5121
H3	FR + FR ² + Season	976	0.6713	0.6703	5077	5102
H4	FR + FR ² + League + Season	976	0.6717	0.6693	5084	5128

Table 21: Table E2. Nested F-tests: extending Model H with League and Season controls.

Comparison	df	F-statistic	p-value
H2 vs H: Adding League FE	4	0.309	0.8721
H3 vs H: Adding Season_Index	1	0.028	0.8660
H4 vs H: Adding League + Season (joint)	5	0.253	0.9385
H4 vs H2: Season beyond League	1	0.029	0.8655
H4 vs H3: League beyond Season	4	0.309	0.8722

- **Model H2:** League_Position ~ FR + FR-squared + League (4 league dummy variables)
- **Model H3:** League_Position ~ FR + FR-squared + Season_Index (linear time trend only)
- **Model H4:** League_Position ~ FR + FR-squared + League + Season_Index (both controls)

All three are compared to the baseline Model H via AIC and formal nested F-tests.

What the results show: Model H4 achieves $R^2 = 0.6714$ and $AIC = 5079.27$, compared to Model H's $R^2 = 0.6709$ and $AIC = 5070.52$. The nested F-test comparing H4 to Model H yields $F(5) = 0.247$, $p = 0.941$, which is **not significant** at $p < 0.05$ — Model H wins on AIC by 8.75 points. Neither the League fixed effects ($F(4) = 0.304$, $p = 0.876$) nor the Season_Index term ($F(1) = 0.023$, $p = 0.880$) add significant explanatory power beyond the quadratic baseline. The Season_Index coefficient in H4 is -0.006 ($SE = 0.036$, $p = 0.879$), confirming no statistically meaningful time trend. None of the four League fixed effects are statistically significant (all $p > 0.35$). **Conclusion: Model H (FR + FR-squared) is the correct and sufficient specification.** Adding league intercepts or a time control does not improve fit — the quadratic Financial Rank model generalises equally well across all five leagues and all ten seasons without further adjustment.

Table 22: Table E3. Model H4 full coefficient table (quadratic FR + League fixed effects + Season trend).

Term	Estimate	Std. Error	t-value	p-value
(Intercept)	0.2022	0.4583	0.4411	0.6592
Financial_Rank	1.2358	0.0768	16.0997	< 0.001 ***
I(Financial_Rank^2)	-0.0206	0.0036	-5.6903	< 0.001 ***
LeagueLa Liga	0.3091	0.3356	0.9209	0.3573
LeagueLigue 1	0.2523	0.3369	0.7491	0.4540
LeaguePremier League	0.3111	0.3356	0.9270	0.3542
LeagueSerie A	0.3055	0.3356	0.9102	0.3630
Season_Index	-0.0062	0.0364	-0.1694	0.8655

Table 23: Model H fit separately on the Early (2015–16 to 2019–20) and Late (2020–21 to 2024–25) halves of the decade.

Period	N	R ²	Adj. R ²	AIC	FR slope	95% CI low	95% CI high	FR ² slope
Early (2015–16 to 2019–20)	490	0.6463	0.6448	2591	1.25	1.029	1.470	-0.0219
Late (2020–21 to 2024–25)	486	0.6975	0.6962	2487	1.21	1.006	1.415	-0.0185

10 Step 7 — Extended Robustness and External Validity

The robustness checks in Step 6 all use the same outcome variable (final league position) and the same pooled dataset. This section pushes further along three independent axes: a cleaner, more intuitive time-split test; a completely different outcome variable (points instead of position); and a completely different sport (the NBA) with a fundamentally different financial structure — a hard salary cap. If Model H’s finding is genuinely robust, it should survive all three.

10.1 A Cleaner Two-Period Robustness Check

Step 6 already showed that a *continuous* FR $\times$ Season_Index interaction is not significant. Here we repeat the idea in the plainest possible way: split the 10 seasons into two five-season halves — **Early** (2015–16 to 2019–20) and **Late** (2020–21 to 2024–25) — and re-fit Model H completely independently on each half. If financial inequality has been increasing (a “super-league effect”) or shrinking (better analytics helping smaller clubs), the fitted curve should visibly differ between the two halves.

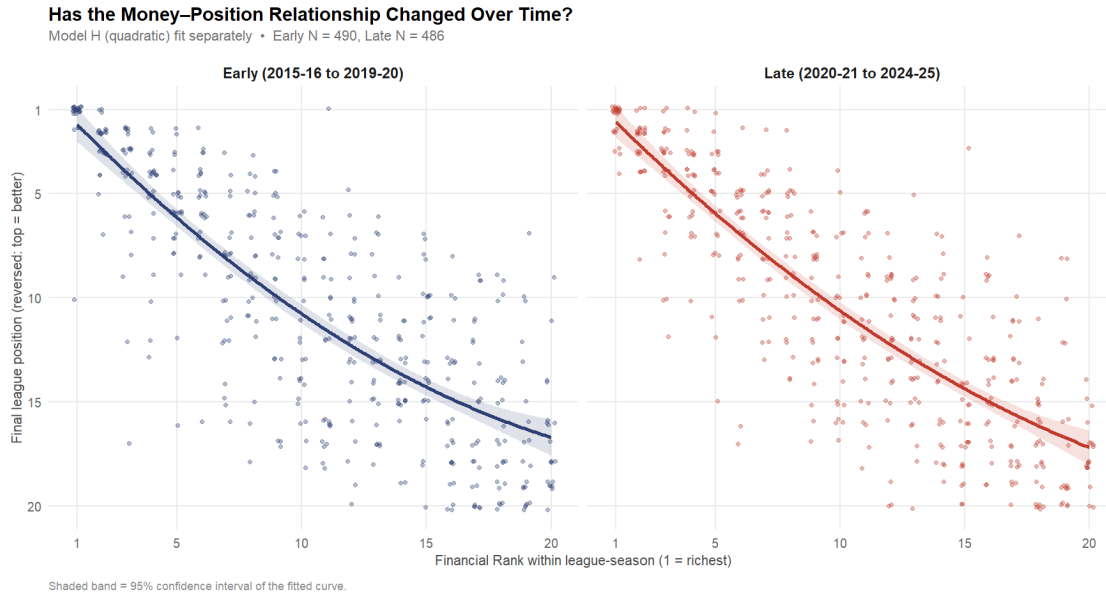


Figure 14: Financial Rank vs. League Position, fit separately for each five-season half. The two quadratic curves are visually almost indistinguishable.

The Early-period FR slope is 1.2497 (95% CI [1.0292, 1.4702]); the Late-period slope is 1.2102 (95% CI [1.0056, 1.4147]) — the intervals overlap almost completely. A formal interaction test (Financial_Rank $\times$ Period, fit on the pooled data) confirms this: the interaction coefficient is -0.0395 and **not statistically significant** ($p = 0.797$).

Independent confirmation of time stability. Two entirely different test designs — a continuous FR $\times$ Season_Index interaction (Step 6) and a binary Early/Late split with its own interaction test (here) —

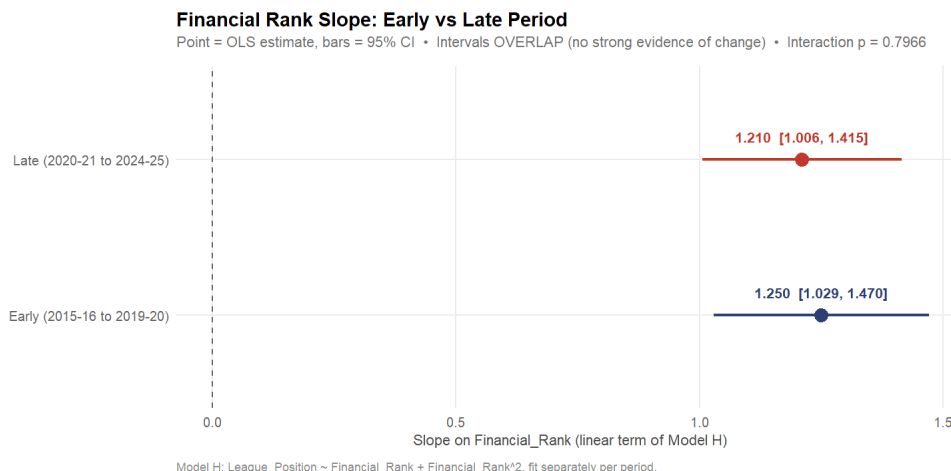


Figure 15: Financial Rank slope with 95% confidence interval, Early vs. Late period. The intervals overlap substantially.

Table 24: Model H re-fit with Points as the outcome: Points FR + FR-squared.

Term	Estimate	Std. Error	t-value	p-value
(Intercept)	85.1162	1.0403	81.82	< 0.001 ***
Financial_Rank	-5.0466	0.2312	-21.83	< 0.001 ***
I(Financial_Rank^2)	0.1268	0.0109	11.67	< 0.001 ***

reach the same conclusion: the financial-to-position relationship has not measurably changed between the first and second half of the decade studied.

10.2 Robustness to the Outcome Measure: Points Instead of Position

Final league position is a *rank*, not a raw quantity — a club that misses out on the title by one point looks identical, position-wise, to one that misses out by twenty. **Points** are a continuous measure of season-long dominance and provide an independent check: does the same financial hierarchy show up when we stop compressing performance into a 1–20 ranking?

As expected, the sign flips relative to the position model: because higher points are *better* (unlike position, where 1st is best), the Financial_Rank coefficient is now **negative** — each step down the financial hierarchy costs a club points, not positions. The model explains $R^2 = 0.678$ of the variation in points, comparable to the $R^2 = 0.671$ obtained with position as the outcome.

Does the outlier ranking change? We recomputed residuals from this points-based model and re-ranked every team-season by absolute residual size.

Leicester City 2015–16 still has the single largest residual — with points as the outcome, exactly as with position. This is a reassuring confirmation: the “greatest statistical upset” finding from Step 5 is not an artefact of using position rather than points.

What *does* change is which clubs round out the rest of the list. Where the position-based residuals (Step 4) foreground overperforming underdogs, the points-based residuals surface a complementary story: **rich clubs that collapsed**. AS Monaco (2018–19 and 2019–20), Chelsea (2015–16 and 2022–23), Bayer Leverkusen (2016–17), and VfL Wolfsburg (2015–16 and 2016–17) all appear among the ten largest points residuals — all financially top-5 clubs that won far fewer points than their wealth predicted. Chelsea’s back-to-back

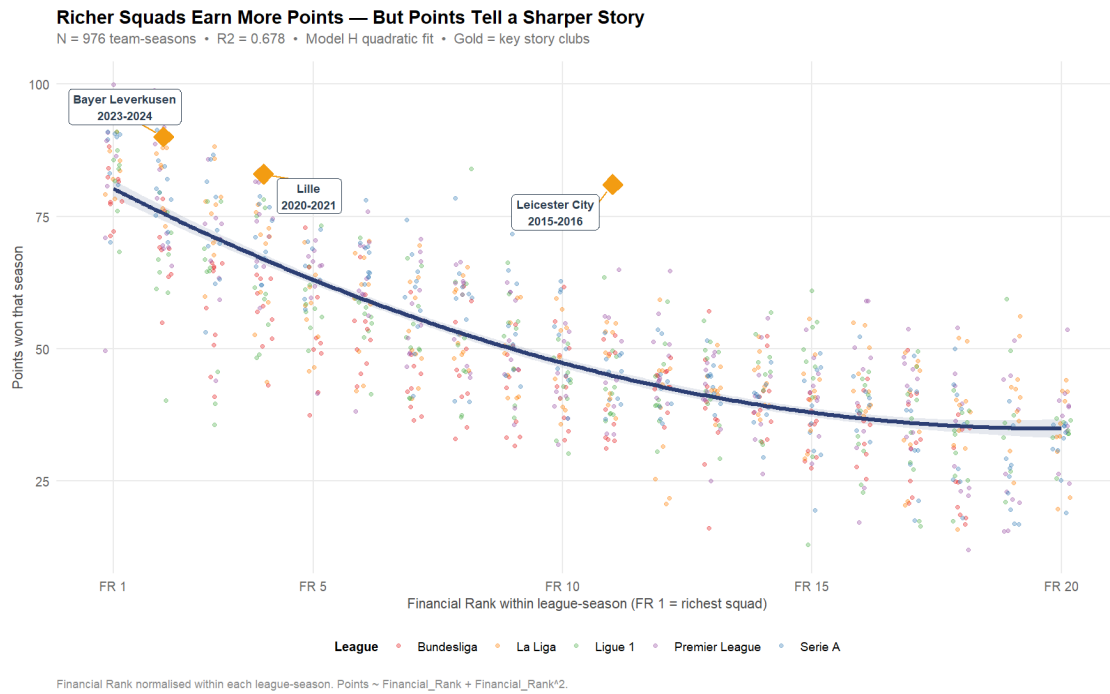


Figure 16: Financial Rank vs. Points won that season, with the quadratic fit overlaid. Gold diamonds mark the three key story clubs from Steps 4–5.

Table 25: Top 10 largest |residual| under the points-based model, alongside each team’s rank in the original position-based model.

Team	Season	Fin. Rank	Points	Points Residual	Rank (position model)
Leicester City	2015-2016	11	81	36.1	3
Lyon	2019-2020	2	40	-35.5	185
AS Monaco	2018-2019	3	36	-35.1	1
Lens	2022-2023	8	84	31.1	38
AS Monaco	2019-2020	3	40	-31.1	114
Chelsea	2015-2016	1	50	-30.2	14
Bayer Leverkusen	2016-2017	3	41	-30.1	15
Chelsea	2022-2023	3	44	-27.1	16
VfL Wolfsburg	2015-2016	3	45	-26.1	195
VfL Wolfsburg	2016-2017	5	37	-26.1	4

Table 26: Football vs. NBA: strength of the money-predicts-finish relationship, 2015–16 to 2024–25.

Sport	N	Pearson r	R^2	AIC	Unique Champions	% Titles: Top-3 Payroll
Football (Top-5 European Leagues)	976	0.8127	0.6713	5075	16	92
NBA	300	0.4364	0.2027	2092	8	50

collapses (defending champions finishing mid-table in 2015–16, then a 12th-place finish in 2022–23 despite record transfer spending under new ownership) and AS Monaco’s post-2017 financial retrenchment are well-known real-world stories that this purely statistical residual ranking recovers without being told to look for them. Lens (2022–23, Financial Rank 8) is the only genuinely new *positive* surprise that does not already appear among the position-model’s largest residuals.

Robust to outcome measure. Whether success is measured by final position or by points won, Financial Rank predicts outcomes with comparable strength, and the single largest statistical anomaly in the dataset — Leicester City 2015–16 — is identified either way.

10.3 External Validity: Does Money Buy Success Outside Football?

Every check so far uses European football data. A stronger test of “does money buy success” is whether the same statistical pattern appears in a completely different sports league — ideally one with a structurally different financial system. The NBA is a natural candidate: unlike Europe’s football leagues, it operates under a **hard salary cap**, deliberately designed (like UEFA’s Financial Fair Play, mentioned in the Introduction) to limit financial dominance. If money still predicts finish as strongly in a capped league as in an uncapped one, that would suggest caps do little to level the field; if the relationship is much weaker, that is evidence the cap is doing its job.

We collected NBA team payroll and final standings for the same ten seasons studied for football (2015–16 to 2024–25), using ESPN’s standings API (final regular-season record, pooled across both conferences and ranked 1–30 by win percentage) and Spotrac’s historical team salary-cap tables (ranked 1–30 by total cap allocations, 1 = highest payroll). We then fit the identical Model H specification: $\text{Final_Standing} \sim \text{Payroll_Rank} + \text{Payroll_Rank}^2$.

A definitional caveat. Unlike European football, where the league table winner *is* the champion, the NBA crowns its champion through a separate playoff tournament — the best regular-season team frequently does not win the title (Oklahoma City Thunder’s 2024–25 championship, for example, came with a payroll rank of only 23rd out of 30). **Final_Standing** below always refers to the regular-season finish (directly comparable to football’s **League_Position**), while “champion” in the summary table below refers to the actual NBA Finals winner each season — a stricter, more literal reading of “champion” than simply topping the table.

The difference is stark. In football, Financial Rank explains $R^2 = 0.671$ of final position ($r = 0.813$, $N = 975$). In the NBA, Payroll Rank explains only $R^2 = 0.203$ of final standing ($r = 0.436$, $N = 300$) — a much noisier relationship, visible directly in the right-hand scatter of the figure above, where high-payroll and low-payroll teams are far more intermixed across the standings. The champion-level statistic tells the same story from a different angle: 92% of European titles went to a financial top-3 club, versus only 50% of NBA titles going to a top-3 payroll team (5 of 10, including the Thunder’s 2024–25 title won with the league’s 23rd-highest payroll — a result that would be a Leicester-level statistical outlier in the football data, but is a comparatively routine occurrence in the NBA).

Money predicts success everywhere, but the salary cap visibly weakens the relationship. The core “money buys success” pattern is not unique to European football — it re-appears in the NBA — but the strength of that pattern is far from universal. A league office can meaningfully weaken the link between spending and winning through plan design (a hard cap), even though it cannot eliminate it entirely. This is indirect but genuine evidence in favour of the logic behind mechanisms like UEFA’s Financial Fair Play, cited as motivation in the Introduction.

Does Money Buy Success? Football vs. the Salary-Capped NBA

Same quadratic specification (Model H) fit separately on each sport — tighter scatter and higher R^2 means money predicts finish more strongly

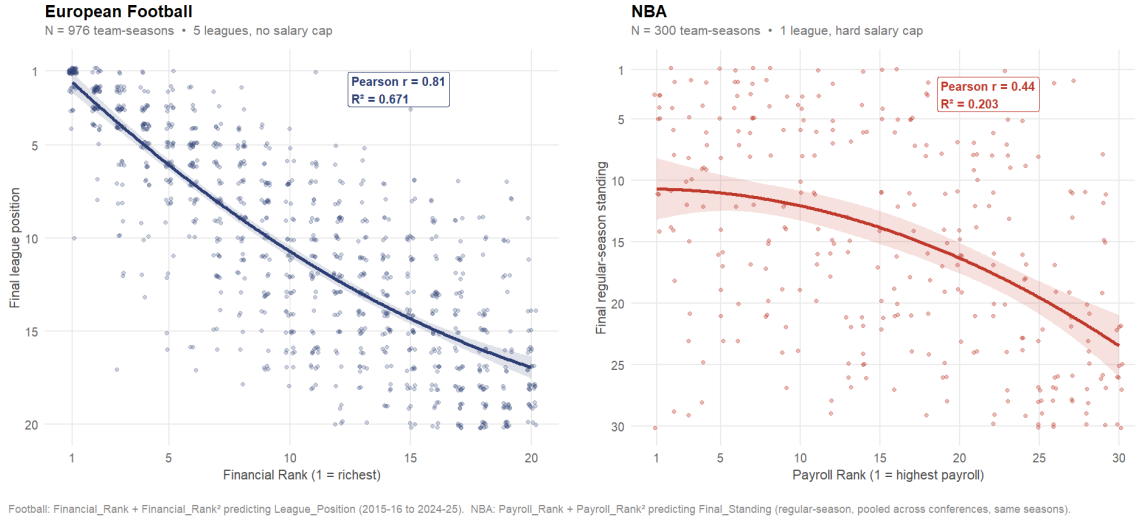


Figure 17: Side-by-side comparison: Financial Rank vs. League Position (European football, no salary cap) and Payroll Rank vs. Final Standing (NBA, hard salary cap). Both panels share the same quadratic specification and axis convention (rank 1 at the top).

Limits of this comparison. The NBA dataset is much smaller (300 vs. 975 team-seasons; only 10 champion-seasons vs. 50), so the champion-level percentages in particular should be read with caution. This is also a single alternative sport with numerous other structural differences from football (playoff-determined titles, a fixed 30-team league, no promotion/relegation, different broadcast and roster-size economics) — any of which, not only the salary cap, could contribute to the weaker relationship observed.

11 Step 8: Refining the Outcome Model with a Generalized Additive Specification

11.1 Choosing a Continuous Outcome: Points Per Game Rather Than League Position

Every specification examined so far, from the linear baseline through Model H's quadratic form, has treated final league position as the outcome to be explained. Position is a natural choice because it is the number supporters and pundits actually reference, but it is not, strictly speaking, well suited to ordinary least squares. League position is an ordinal ranking: it tells us that a team finished third and not fourth, but it says nothing about the size of the gap between them. Two adjacent positions can represent a single point separating two closely matched sides in one season, and a gap of twenty points in another. Treating an ordinal variable as though it were measured on an interval scale is a simplification, and one that a genuinely continuous outcome avoids entirely. Points per game aggregates every match result into a single continuous quantity, is directly comparable across leagues of different sizes and across the one shortened season in the sample, and removes the implicit assumption that the distance between first and second place equals the distance between nineteenth and twentieth. Where an ordinal outcome such as position is better suited to rank based methods, a continuous outcome such as points per game fits naturally within the regression framework already used throughout this report, and it is the outcome adopted for the remainder of this analysis.

11.2 From a Fixed Quadratic Term to a Smooth Term

The quadratic specification used throughout Steps 2 through 7 (Financial Rank plus Financial Rank squared) was itself a considerable improvement over the linear baseline: it captured the sense that the very richest clubs are disproportionately dominant, a pattern a straight line cannot represent. A quadratic term is, however, still a strong assumption about the shape of that relationship. It forces the curve connecting financial rank to performance into the specific shape of a parabola, with a single point of inflection and symmetric curvature on either side. There is no substantive reason to expect the true relationship between financial rank and points per game to take exactly this form, rather than, for example, levelling off gradually among mid table clubs before steepening again near the bottom of the table. A generalized additive model relaxes this assumption. Instead of fitting a fixed polynomial chosen in advance, a GAM fits a smooth term (a flexible curve built from a basis of simpler functions) and lets the data determine how much, and where, that curve should bend. The specification fitted here is points per game as a smooth function of financial rank, estimated with the `mgcv` package in R using restricted maximum likelihood smoothing parameter selection. This preserves the intuition behind the quadratic term, that the effect of financial rank is nonlinear, while removing the requirement that the nonlinearity take one specific, predetermined shape.

11.3 Comparing the Three Specifications

The table below compares the plain linear specification, the original quadratic Model H, and the new generalized additive model, all fitted with points per game as the outcome and financial rank as the sole predictor. For the linear and quadratic rows the fit statistic is R squared; for the generalized additive model, whose predictions are not produced by a simple sum of fixed coefficients, the comparable quantity is deviance explained, and effective degrees of freedom is reported only where a smooth term is present.

Table 27: Comparison of the linear, quadratic, and generalized additive specifications for points per game. Source: `extensions/outputs/gam_revision/gam_revision_table.csv`.

Model	Outcome	N	R squared / deviance explained	AIC	Effective degrees of freedom
Linear	Points per game	976	0.652	228.50	N/A
Quadratic (Model H)	Points per game	976	0.696	100.10	N/A
Generalized additive model	Points per game	976	0.707	75.32	6.30

The ordering is consistent across both fit statistics. Moving from a straight line to a fixed quadratic term already raises the explained variation from 0.652 to 0.696 and lowers AIC by well over a hundred points; moving from the quadratic term to a smooth term raises the explained variation further, to a deviance explained of 0.707, and lowers AIC again, to 75.32. Both nonlinear specifications comfortably outperform the linear model, and the generalized additive model in turn outperforms the quadratic one on both statistics at once.

The effective degrees of freedom column offers a direct way to judge how much curvature the model actually needed. An effective degrees of freedom close to one would indicate that the smooth term is behaving almost exactly like a straight line, in which case the added flexibility of a GAM would have bought nothing. Here, the effective degrees of freedom is 6.3, well above one, which confirms genuine, nontrivial curvature in the relationship between financial rank and points per game that a straight line would miss entirely. At the same time, an effective degrees of freedom of 6.3 is also more nuanced than the single bend a fixed quadratic

term allows: a quadratic curve can only bend in one place, in one direction, whereas a smooth term of this flexibility can accommodate a more gradual, multi part curve, including a steep early drop among the wealthiest clubs and a flatter tail among poorer ones. In short, the data support meaningfully more curvature than a parabola allows, without requiring anywhere near the flexibility of an unconstrained, highly wiggly curve.

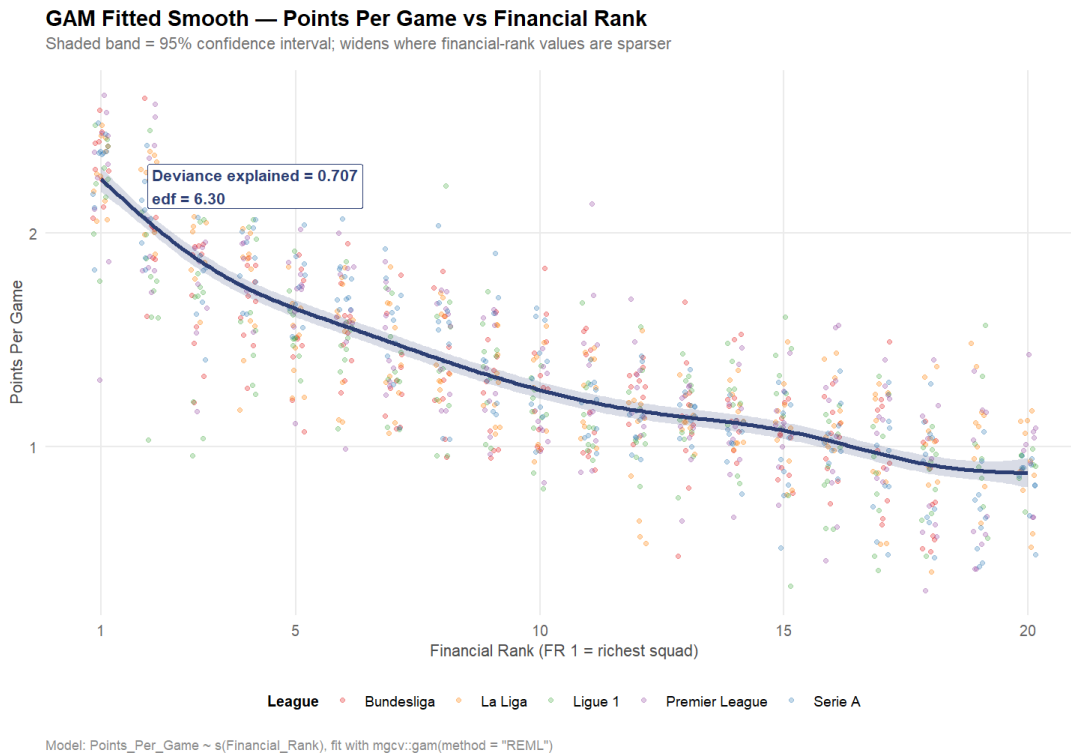


Figure 18: The GAM’s fitted smooth term (dark line) and 95 percent confidence band (shaded) plotted against financial rank, with individual team seasons jittered by league in the background.

The fitted curve, shown above, illustrates this well supported curvature directly: points per game falls steeply as financial rank worsens from first to roughly tenth, then flattens into a shallower decline for the remaining, less wealthy clubs. One limitation is visible in the same figure. The confidence band is narrow through most of the financial rank range, but it widens noticeably for financial ranks above approximately 15. This follows directly from data availability rather than any weakness in the smoothing procedure: because the Bundesliga fields only eighteen clubs per season, financial ranks of 19 and 20 are populated exclusively by the other four leagues, each of which contributes twenty clubs. The result is a genuinely smaller sample at the poorest end of the financial rank scale, and the wider band there is an honest reflection of that reduced sample size rather than evidence that the model performs worse for those clubs.

11.4 Robustness of the Central Finding

A natural concern when switching models is whether the substantive story changes along with the statistics. It does not. Recomputing residuals under the generalized additive model and reapplying the same 1.5 standard deviation threshold used throughout this report, Leicester City’s 2015-16 title winning season remains the single largest overperformance in the dataset, with a residual of +0.921 points per game, larger in magnitude than under the linear specification. Bayer Leverkusen’s unbeaten 2023-24 championship also remains a clear overperformer, at +0.598 points per game. Both clubs continue to stand out as statistical anomalies regardless of whether the outcome is league position, points, or points per game, and regardless of

whether the functional form connecting financial rank to performance is linear, quadratic, or an unconstrained smooth term. The overall residual spread also tightens under the new specification, from 0.271 points per game under the linear model to 0.249 under the generalized additive model, a further sign that the smooth term is capturing genuine structure in the data rather than simply adding flexibility for its own sake.

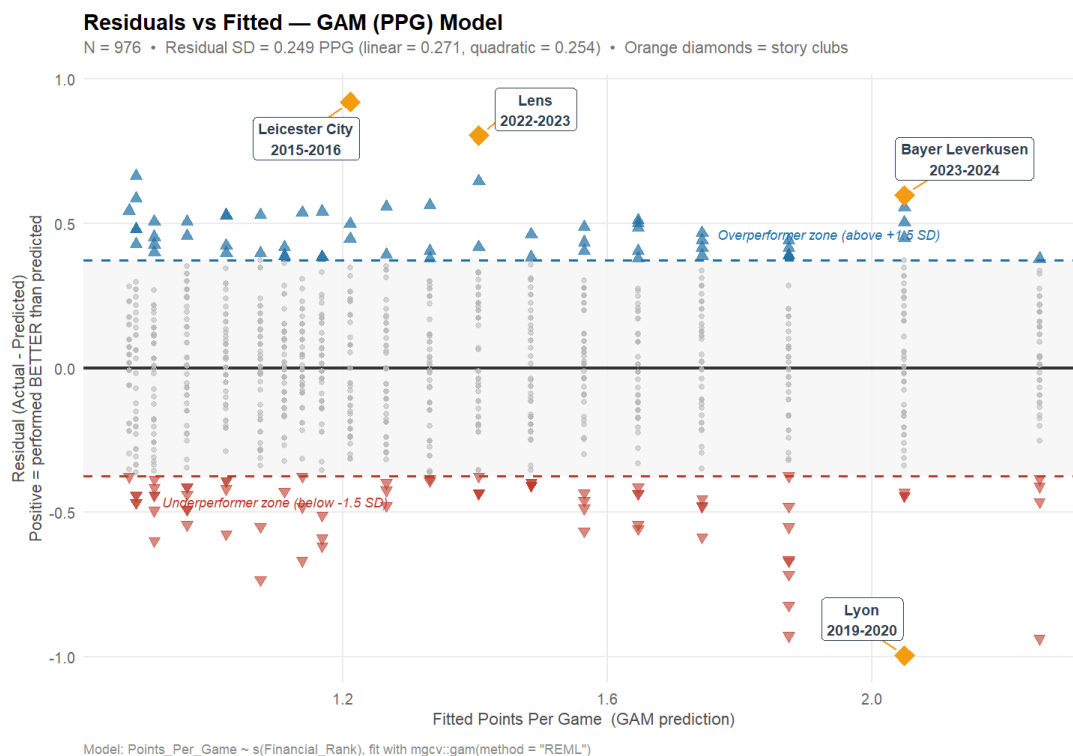


Figure 19: Residuals versus fitted values under the generalized additive model, using the same overperformer and underperformer zones (1.5 standard deviations) as the earlier residual plots in this report.

The complete R code used to fit these models, compute their diagnostics, and produce the accompanying figures is provided in Appendix E.

Given its stronger fit on both deviance explained and AIC, its more defensible statistical assumptions, and its confirmation of every substantive finding established under the earlier specifications, the generalized additive model fitted to points per game is adopted as the primary model for the remainder of this analysis.

12 Conclusions

12.1 What We Found: A Summary

This study has systematically examined whether financial strength predicts sporting success in Europe's five major football leagues over 10 seasons, and then stress-tested that answer against alternative time periods, an alternative outcome measure, and an entirely different sport. The answer is unambiguous: **yes, and very strongly so**. Here are the six main findings:

Finding 1 — Money very strongly predicts where clubs finish. The Pearson correlation between Financial Rank and final league position is $r = 0.813$ (Spearman $= 0.814$), both statistically significant

at $p < 0.001$ across all 976 team-seasons. The best-fitting model (Model H, quadratic in FR) explains $R^2 = 0.671$ — nearly two-thirds of all variation in final standings — using nothing but a club’s within-league wealth rank.

Finding 2 — The relationship is non-linear: diminishing returns at the bottom. Adding a squared term (FR²) significantly improves model fit ($F = 31.71$, $p < 0.001$). The marginal positional penalty of dropping one rank is largest at the top of the financial hierarchy (~1.19 positions per rank at FR 1→2) and shrinks toward the bottom (~0.63 positions per rank at FR 15→16). In plain language: the gap between the richest and second-richest club is much larger than the gap between the 15th and 16th richest.

Finding 3 — The effect is the same in every league and has not changed over time. The FR × League interaction is not significant ($F = 1.17$, $p = 0.32$), meaning the financial hierarchy operates identically in Germany, Spain, France, England, and Italy. The FR × Season time trend is also not significant ($F = 0.44$, $p = 0.51$), meaning the relationship has been equally strong throughout 2015–16 to 2024–25. Financial dominance is a universal and stable feature of European football, not a recent trend or a league-specific phenomenon.

Finding 4 — Exceptions exist, but only one is truly extraordinary. Of 50 league titles across the decade, only 4 were won by clubs outside the financial top three. Three of those four (Napoli, Lille, and Liverpool) were at FR 4–5, barely outside the expected zone, and within one standard deviation of expectation. Only **Leicester City 2015–16** — Financial Rank 11, predicted to finish 11th, actually won the title — is a genuine statistical outlier (-3.2 SDs from the mean, highest Cook’s D in the dataset). Such events are statistically exceptional and cannot be used to argue that money does not matter.

Finding 5 — The result survives a battery of independent robustness checks. Splitting the decade into two independent five-season halves (above) yields statistically indistinguishable Financial Rank slopes (interaction $p = 0.797$); re-running Model H with **points** instead of position as the outcome (above) still identifies Leicester City 2015–16 as the single largest statistical anomaly, while additionally surfacing a complementary story of financially strong clubs (Chelsea, AS Monaco, Bayer Leverkusen) that badly underspent their points total relative to expectation. Neither check weakens the core finding.

Finding 6 — The pattern generalises beyond football, but a salary cap visibly weakens it. The same quadratic specification fit on ten seasons of NBA data (above) confirms that money-predicts-finish is not football-specific: Payroll Rank significantly predicts NBA final standing. But the relationship is far weaker under the NBA’s hard salary cap ($R^2 = 0.203$, $r = 0.436$) than in football’s unrestricted spending environment ($R^2 = 0.671$, $r = 0.813$), and only half of NBA titles in the period went to a top-3 payroll team, versus 92% in football. This is indirect evidence that deliberate cap-style mechanisms can meaningfully loosen — without eliminating — the link between spending and winning.

12.2 Limitations and Future Directions

Every study has limitations, and being transparent about them is part of good scientific practice. The table below summarises the main limitations and suggests how future research could address them.

13 Technical Appendix

13.1 A. Full Coefficient Table (All Models A–E)

```
coef_all <- read_csv("outputs/08_regression_results.csv", show_col_types = FALSE)
```

Table 28: Study limitations and suggested future extensions.

Limitation	Issue	Potential Fix
Market value as quality proxy	Transfermarkt valuations may lag actual squad quality by weeks; subjective editor estimates	Use wage bill data as an alternative or cross-validate with Opta squad ratings
No managerial quality variable	Exceptional coaches (Ranieri at Leicester) are not captured in the financial model	Add manager tenure, prior record, or a coaching quality index
No injury data	Injury crises can dramatically alter a squad's effective quality mid-season	Incorporate injury-adjusted squad value (e.g., weighted by availability %)
Panel structure ignored	Teams appear multiple seasons; correlated errors across seasons are unmodelled	Fixed-effects panel model or clustered standard errors at the team level
Top divisions only	Results may not generalise to lower divisions or smaller national leagues	Replicate in Eredivisie, Championship, Scottish Premiership
NBA comparison is a single, small sample	The NBA cross-sport comparison above rests on only 300 team-seasons and 10 champion-seasons; NBA titles are also playoff-determined, not table position, making the sample smaller and structurally different from the football data	Extend to further capped/uncapped leagues (NFL, MLB, EuroLeague basketball) for a larger cross-sport sample

```

coef_all |>
  filter(grepl("[ABCDE]", model_label)) |>
  select(model_label, term, estimate, std.error, statistic, p.value) |>
  mutate(
    across(c(estimate, std.error, statistic), ~ round(., 4)),
    p_display = case_when(
      p.value < 0.001 ~ "< 0.001 ***",
      p.value < 0.01 ~ sprintf("%.4f **", p.value),
      p.value < 0.05 ~ sprintf("%.4f *", p.value),
      TRUE ~ sprintf("%.4f", p.value)
    ),
    model_label = sub(":.*", "", model_label)
  ) |>
  select(-p.value) |>
  kbl(longtable = TRUE,
    col.names = c("Model", "Term", "Estimate", "Std. Error",
                  "t / z", "p-value"),
    caption = "Table A1. Full regression coefficient table (Models A-E).")
  ) |>
  kable_styling(
    latex_options = c("repeat_header", "striped"),
    full_width = FALSE
  ) |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white") |>
  collapse_rows(columns = 1, valign = "top")

```

Table 29: Table A1. Full regression coefficient table (Models A–E).

Model	Term	Estimate	Std. Error	t / z	p-value
A	(Intercept)	1.9278	0.2190	8.8038	< 0.001 ***
A	Financial_Rank	0.8127	0.0187	43.5364	< 0.001 ***
B	(Intercept)	1.7875	0.3040	5.8795	< 0.001 ***
B	Financial_Rank	0.8118	0.0187	43.3162	< 0.001 ***
B	LeagueLa Liga	0.1922	0.3402	0.5650	0.5722
B	LeagueLigue 1	0.1577	0.3417	0.4617	0.6444
B	LeaguePremier League	0.1922	0.3402	0.5650	0.5722
B	LeagueSerie A	0.1882	0.3402	0.5531	0.5803
C	(Intercept)	2.8889	0.2539	11.3798	< 0.001 ***
C	Log_Normalized_Value	-5.0289	0.1508	-33.3508	< 0.001 ***
D	(Intercept)	-0.2152	0.4049	-0.5314	0.5953
D	Log_Normalized_Value	-5.9663	0.1439	-41.4671	< 0.001 ***
D	LeagueLa Liga	-0.2012	0.3507	-0.5736	0.5664
D	LeagueLigue 1	-1.1028	0.3542	-3.1133	0.0019 **
D	LeaguePremier League	4.0859	0.3573	11.4346	< 0.001 ***
D	LeagueSerie A	2.7888	0.3521	7.9194	< 0.001 ***
D	Season_Index	0.1059	0.0381	2.7781	0.0056 **
E	(Intercept)	1.6416	0.4033	4.0701	< 0.001 ***
E	Financial_Rank	-1.2985	0.1900	-6.8343	< 0.001 ***

13.2 B. Model H Quadratic Coefficients (Detailed)

```
coef_H_raw |>
  mutate(
    term = recode(term,
      "(Intercept)"      = "Intercept",
      "Financial_Rank"    = "Financial Rank (b1)",
      "I(Financial_Rank^2)" = "FR-squared (b2)"
    ),
    across(c(estimate, std.error, statistic, conf.low, conf.high), ~ round(., 5)),
    p.value = formatC(p.value, format = "e", digits = 3)
  ) |>
  select(term, estimate, std.error, statistic, p.value, conf.low, conf.high) |>
  kbl(longtable = FALSE,
    col.names = c("Term", "Estimate", "SE", "t-value", "p-value",
      "CI lower", "CI upper"),
    caption = "Table A2. Model H (Position ~ FR + FR-sq) full coefficient table with 95pct CIs."
  ) |>
  kbs() |>
  row_spec(0, bold = TRUE, background = "#1A2A4A", color = "white")
```

13.3 C. R Code Used in This Analysis

The complete R scripts used to produce all data, models, and outputs are available in the project directory. Below we reproduce the key analysis steps for transparency and reproducibility.

Data loading and model setup:

Table 30: Table A2. Model H (Position + FR + FR-sq) full coefficient table with 95pct CIs.

Term	Estimate	SE	t-value	p-value	CI lower	CI upper
Intercept	0.4172	0.3441	1.212	2.257e-01	-0.2573	1.0916
Financial Rank (b1)	1.2309	0.0765	16.093	7.648e-52	1.0810	1.3808
FR-squared (b2)	-0.0203	0.0036	-5.633	2.323e-08	-0.0273	-0.0132

```
# Load and clean the master dataset (corrected: SPAL 2017-18 market value fix)
df <- read_csv("data/merged/master_dataset_corrected.csv", show_col_types = FALSE) |>
  filter(!is.na(League_Position), !is.na(Financial_Rank),
         !is.na(Market_Value_M_EUR), !is.na(Log_Normalized_Value)) |>
  mutate(
    League = factor(as.character(League), levels = league_order),
    Champion_bin = as.integer(
      tolower(as.character(Champion)) %in% c("yes", "true", "1")
    )
  )
```

Fitting the regression models:

```
# OLS models
modA <- lm(League_Position ~ Financial_Rank, data = df)
modB <- lm(League_Position ~ Financial_Rank + League, data = df)
modC <- lm(League_Position ~ Log_Normalized_Value, data = df)
modD <- lm(League_Position ~ Log_Normalized_Value + League + Season_Index, data = df)
modH <- lm(League_Position ~ Financial_Rank + I(Financial_Rank^2), data = df)

# Logistic regression for championship probability
modE <- glm(Champion_bin ~ Financial_Rank, data = df, family = binomial())
```

Computing residuals and outlier flags:

```
aug_H <- augment(modH, data = df) |>
  rename(Fitted_H = .fitted, Residual_H = .resid, CooksD = .cooksD)

resid_sd <- sd(aug_H$Residual_H, na.rm = TRUE)

aug_H <- aug_H |>
  mutate(
    Outlier_Flag = case_when(
      Residual_H < -1.5 * resid_sd ~ "Overperformer",
      Residual_H > 1.5 * resid_sd ~ "Underperformer",
      TRUE ~ "Normal"
    )
  )
```

Model comparison using AIC:

```
# Compare all OLS models by AIC, R-squared, and adjusted R-squared
model_comparison <- list(A = modA, B = modB, C = modC, D = modD, H = modH) |>
  map_dfr(glance, .id = "Model") |>
```

R version	4.5.3
Platform	Windows 11 x64 (build 26200)
Date knitted	2026-07-11
Key packages	dplyr, ggplot2, broom, kableExtra, ggrepel

```
transmute(Model, R2 = round(r.squared, 4),
  Adj_R2 = round(adj.r.squared, 4),
  AIC     = round(AIC, 1),
  BIC     = round(BIC, 1))
```

Key analysis scripts in the project:

Script	Purpose
RScripts/01_scrape_ESPN_standings.R	Download league standings from ESPN
RScripts/02_scrape_transfermarkt_market_values.R	Download squad market values
RScripts/03_compute_financial_rank.R	Assign within-league Financial Rank
RScripts/04_merge_standings_and_market_values.R	Build master dataset
RScripts/05_analysis_and_regression.R	Core OLS and logit models
RScripts/05b_extended_regression.R	Robustness checks and extended models
RScripts/05c_model_h4_extended.R	Model H4: League + Season extensions (Section 9.4)
RScripts/06_residuals_and_outliers.R	Residual analysis, Cook's Distance
RScripts/07_beautiful_charts.R	Publication-quality visualisations

13.4 D. Session Information

13.5 E. R Code for the Points-Per-Game GAM Revision (Step 8)

The linear and generalized additive models presented in Step 8, along with their comparison tables and diagnostic plots, are produced by a single self-contained script. The complete, unmodified contents of `extensions/scripts/16_linear_gam_revision.R` are reproduced below for transparency and reproducibility; the chunk is not evaluated when this report is knitted.

```
# =====
# 16_linear_gam_revision.R
# Responds to Jacob's updated feedback: league position is not a
# suitable outcome (not continuous), and polynomial terms should
# not be used. This drops the quadratic Model H entirely and tests
# two alternatives, both using Points_Per_Game as the outcome and
# Financial_Rank as the only predictor:
#
#   Model 1 - Linear:  Points_Per_Game ~ Financial_Rank
#   Model 2 - GAM:    Points_Per_Game ~ s(Financial_Rank)
#
# Uses data/merged/master_dataset_corrected.csv (SPAL 2017-18 Serie A
# market value fix from extensions/scripts/15_spal_correction.R) -
# run script 15 first if that file doesn't exist yet.
#
# Table 1 - This script's two new models vs the quadratic Model H
#           (PPG) reference, read from E1_model_evolution_table.csv
```



```

# Table 2 - Combined master comparison across the whole project:
#           all four E1 rows (read-only) + the two new models here,
#           flagged against Jacob's feedback criteria
# Plot 1 - Residuals vs fitted for the linear PPG model (E5 style)
# Plot 2 - GAM fitted smooth vs Financial_Rank with CI band
# Plot 3 - Residuals vs fitted for the GAM PPG model (E5 style),
#           with a check for new standout outliers vs earlier work
#
# This script only READS extensions/outputs/E1_model_evolution_table.csv
# - it never writes to it. Nothing in extensions/outputs/ (E1-E6) or
# anywhere else in the project is modified.
#
# Outputs (all new, all in extensions/outputs/gam_revision/):
#   gam_revision_table.csv
#   gam_revision_table.html           (formatted version, recommended row bolded)
#   combined_model_comparison.csv
#   combined_model_comparison.html    (formatted version, recommended row bolded)
#   linear_ppg_residuals.png
#   gam_smooth_confidence.png
#   gam_ppg_residuals.png
# =====

source("utils/packages.R")
source("utils/constants.R")
source("utils/helpers.R")
load_packages(c("dplyr", "readr", "tidyr", "ggplot2", "broom", "scales", "purrr", "stringr"))

ensure_installed <- function(pkgs) {
  missing <- pkgs[!sapply(pkgs, requireNamespace, quietly = TRUE)]
  if (length(missing) > 0) {
    cat("Installing:", paste(missing, collapse = ", "), "\n")
    install.packages(missing,
                     repos = "https://packagemanager.posit.co/cran/latest",
                     quiet = TRUE)
  }
  invisible(lapply(pkgs, library, character.only = TRUE))
}
ensure_installed(c("ggrepel", "mgcv", "kableExtra"))

OUTPUT_DIR <- "extensions/outputs/gam_revision"
make_dir(OUTPUT_DIR)
OUTPUT_FILES <- character(0)
track <- function(path) { OUTPUT_FILES <- c(OUTPUT_FILES, path); invisible(path) }

theme_clean <- theme_minimal(base_size = 12) +
  theme(
    plot.title       = element_text(face = "bold", size = 14, margin = margin(b = 4)),
    plot.subtitle    = element_text(size = 10.5, colour = "grey45", margin = margin(b = 12)),
    plot.caption     = element_text(size = 8.5, colour = "grey55", hjust = 0, margin = margin(t = 10)),
    axis.title       = element_text(size = 10.5, colour = "grey30"),
    axis.text        = element_text(size = 9.5, colour = "grey40"),
    panel.grid.major = element_line(colour = "grey92", linewidth = 0.5),
    panel.grid.minor = element_blank(),

```

```

    legend.position = "bottom",
    legend.title    = element_text(size = 9.5, face = "bold"),
    legend.text     = element_text(size = 9),
    plot.background = element_rect(fill = "white", colour = NA),
    panel.background = element_rect(fill = "white", colour = NA),
    plot.margin     = margin(16, 16, 12, 16)
  )

league_colours <- c(
  "Bundesliga"      = "#E41A1C",
  "La Liga"         = "#FF7F00",
  "Ligue 1"         = "#4DAF4A",
  "Premier League"  = "#984EA3",
  "Serie A"         = "#377EB8"
)

BIG4_LEAGUES <- c("Premier League", "La Liga", "Serie A", "Ligue 1")

#
# PHASE 0: Load master dataset & derive Points_Per_Game
#
cat("=== PHASE 0: Load Master Dataset ===\n")

master_path <- "data/merged/master_dataset_corrected.csv"
if (!file.exists(master_path)) stop("Cannot find master_dataset_corrected.csv - run extensions/scripts/

master <- read_csv(master_path, show_col_types = FALSE) |>
  mutate(League = as.character(League))

master <- master |>
  mutate(
    Matches_Played = case_when(
      League %in% BIG4_LEAGUES ~ 38,
      League == "Bundesliga"   ~ 34,
      TRUE                     ~ NA_real_
    ),
    Points_Per_Game = Points / Matches_Played
  )

model_df_ppg <- master |>
  filter(!is.na(Financial_Rank), !is.na(Points_Per_Game))

cat(sprintf(" Rows: %d | Complete cases for FR + PPG: %d\n", nrow(master), nrow(model_df_ppg)))
cat("PHASE 0 complete.\n\n")

#
# PHASE 1: Fit the two new models
#
cat("=== PHASE 1: Fit Linear and GAM Models (PPG ~ Financial_Rank) ===\n")

lin_ppg <- lm(Points_Per_Game ~ Financial_Rank, data = model_df_ppg)

```

```

gam_ppg <- mgcv::gam(Points_Per_Game ~ s(Financial_Rank), data = model_df_ppg, method = "REML")

lin_summary <- summary(lin_ppg)
gam_summary <- summary(gam_ppg)

gam_edf      <- unname(gam_summary$s.table[1, "edf"])
gam_devepl   <- gam_summary$dev.expl

cat(sprintf("  Linear PPG model:  R² = %.4f |  AIC = %.2f\n", lin_summary$r.squared, AIC(lin_ppg)))
cat(sprintf("  GAM PPG model:    Deviance explained = %.4f |  edf = %.3f |  AIC = %.2f\n",
            gam_devepl, gam_edf, AIC(gam_ppg)))
cat("PHASE 1 complete.\n\n")

#
# TASK 1 - Table 1: this script's two new models vs quadratic
#           Model H (PPG) reference, read from E1 (not refit)
#   + gam_revision_table.csv
#
cat("=== TASK 1: gam_revision_table (this script's models) ===\n")

e1_path <- "extensions/outputs/E1_model_evolution_table.csv"
if (!file.exists(e1_path)) stop("Cannot find E1_model_evolution_table.csv - run extensions/scripts/13_m")
e1_table <- read_csv(e1_path, show_col_types = FALSE)

modH_ppg_ref <- e1_table |> filter(Model == "H: Quadratic (PPG)")
if (nrow(modH_ppg_ref) != 1) stop("Expected exactly one 'H: Quadratic (PPG)' row in E1_model_evolution_")

gam_revision_table <- tibble(
  Model          = c("H: Quadratic (PPG) [reference]", "Linear (PPG)", "GAM (PPG)"),
  Outcome        = "Points_Per_Game",
  N              = c(modH_ppg_ref$N, nrow(model_df_ppg), nrow(model_df_ppg)),
  R2_or_Dev_Explained = c(modH_ppg_ref$R_squared,
                          round(lin_summary$r.squared, 4),
                          round(gam_devepl, 4)),
  AIC             = c(modH_ppg_ref$AIC, round(AIC(lin_ppg), 2), round(AIC(gam_ppg), 2)),
  EDF            = c(NA_real_, NA_real_, round(gam_edf, 3))
)

path_gam_table <- file.path(OUTPUT_DIR, "gam_revision_table.csv")
write_csv(gam_revision_table, path_gam_table)
track(path_gam_table)
cat(sprintf("  Saved: %s\n", path_gam_table))
print(as.data.frame(gam_revision_table))
cat("TASK 1 complete.\n\n")

#
# TASK 2 - Table 2: combined master comparison across the
#           whole project (E1 rows read-only + this script's rows)
#   + combined_model_comparison.csv
#
cat("=== TASK 2: combined_model_comparison (whole-project master table) ===\n")

```

```

e1_rows <- e1_table |>
  transmute(
    Model,
    Outcome,
    N,
    R2_or_Dev_Explained = R_squared,
    AIC,
    EDF = NA_real_,
    Has_Polynomial_Term = grepl("Quadratic", Model)
  )

new_rows <- tibble(
  Model = c("Linear (PPG)", "GAM (PPG)"),
  Outcome = "Points_Per_Game",
  N = c(nrow(model_df_ppg), nrow(model_df_ppg)),
  R2_or_Dev_Explained = c(round(lin_summary$r.squared, 4), round(gam_devexpl, 4)),
  AIC = c(round(AIC(lin_ppg), 2), round(AIC(gam_ppg), 2)),
  EDF = c(NA_real_, round(gam_edf, 3)),
  Has_Polynomial_Term = c(FALSE, FALSE)
)

combined_table <- bind_rows(e1_rows, new_rows) |>
  mutate(
    meets_feedback_criteria = (Outcome == "Points_Per_Game") & !Has_Polynomial_Term
  ) |>
  select(-Has_Polynomial_Term)

best_fit <- combined_table |>
  filter(meets_feedback_criteria) |>
  slice_max(order_by = R2_or_Dev_Explained, n = 1, with_ties = FALSE)

combined_table <- combined_table |>
  mutate(recommended = meets_feedback_criteria & (Model == best_fit$Model))

path_combined <- file.path(OUTPUT_DIR, "combined_model_comparison.csv")
write_csv(combined_table, path_combined)
track(path_combined)
cat(sprintf(" Saved: %s\n", path_combined))
print(as.data.frame(combined_table))

cat(sprintf(
  "\n SUMMARY: Recommended model = \"%s\" (Points_Per_Game outcome, no polynomial term, %s = %.4f - th",
  best_fit$Model,
  ifelse(best_fit$Model == "GAM (PPG)", "deviance explained", "R-squared"),
  best_fit$R2_or_Dev_Explained
))
cat("TASK 2 complete.\n\n")

#
# TASK 3 - Formatted (HTML) versions with the recommended row bolded
#
cat("=== TASK 3: Formatted HTML Tables ===\n")

```

```

bold_gam_row <- which(gam_revision_table$Model == "GAM (PPG)")
gam_html <- kableExtra::kbl(gam_revision_table, digits = 4,
                           caption = "Table 1: Linear & GAM PPG models vs quadratic Model H (reference)",
                           kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE),
                           kableExtra::row_spec(bold_gam_row, bold = TRUE, background = "#EAF2FB"))
path_gam_html <- file.path(OUTPUT_DIR, "gam_revision_table.html")
kableExtra::save_kable(gam_html, path_gam_html)
track(path_gam_html)
cat(sprintf("  Saved: %s\n", path_gam_html))

bold_combined_row <- which(combined_table$recommended)
combined_html <- kableExtra::kbl(combined_table, digits = 4,
                                 caption = "Table 2: Combined model comparison across the whole project",
                                 kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE),
                                 kableExtra::row_spec(bold_combined_row, bold = TRUE, background = "#EAF2FB"))
path_combined_html <- file.path(OUTPUT_DIR, "combined_model_comparison.html")
kableExtra::save_kable(combined_html, path_combined_html)
track(path_combined_html)
cat(sprintf("  Saved: %s\n", path_combined_html))
cat("TASK 3 complete.\n\n")

#
# TASK 4 - Residuals vs fitted, linear PPG model (E5 style)
#   + linear_ppg_residuals.png
#
cat("=== TASK 4: Linear PPG Residuals vs Fitted ===\n")

aug_lin <- augment(lin_ppg, data = model_df_ppg) |>
  rename(Fitted_PPG = .fitted, Residual_PPG = .resid)

resid_sd_lin <- sd(aug_lin$Residual_PPG, na.rm = TRUE)
overp_thr_lin <- 1.5 * resid_sd_lin
underp_thr_lin <- -1.5 * resid_sd_lin

aug_lin <- aug_lin |>
  mutate(
    Outlier_Flag = case_when(
      Residual_PPG > overp_thr_lin ~ "Overperformer",
      Residual_PPG < underp_thr_lin ~ "Underperformer",
      TRUE ~ "Normal"
    )
  )

cat(sprintf("  Residual SD: %.3f PPG | Overp. threshold: > %.3f | Underp. threshold: < %.3f\n",
            resid_sd_lin, overp_thr_lin, underp_thr_lin))

label_df_lin <- aug_lin |>
  filter(
    (grepl("Leicester", Team) & Season == "2015-2016") |
    (grepl("Leverkusen", Team) & Season == "2023-2024")
  ) |>
  mutate(Label = paste0(Team, "\n", Season))

```

```

leicester_rank_lin <- aug_lin |> arrange(desc(Residual_PPG)) |>
  mutate(Rank = row_number()) |> filter(grepl("Leicester", Team), Season == "2015-2016")
if (nrow(leicester_rank_lin) == 1) {
  cat(sprintf(" Leicester City 2015-16 residual rank under the LINEAR PPG model: %d of %d | Residual_PPG: %d",
    leicester_rank_lin$Rank, nrow(aug_lin), leicester_rank_lin$Residual_PPG))
}
leverkusen_rank_lin <- aug_lin |> arrange(desc(Residual_PPG)) |>
  mutate(Rank = row_number()) |> filter(grepl("Leverkusen", Team), Season == "2023-2024")
if (nrow(leverkusen_rank_lin) == 1) {
  cat(sprintf(" Bayer Leverkusen 2023-24 residual rank under the LINEAR PPG model: %d of %d | Residual_PPG: %d",
    leverkusen_rank_lin$Rank, nrow(aug_lin), leverkusen_rank_lin$Residual_PPG))
} else {
  cat("\n")
}

bg_df_lin <- aug_lin |> filter(Outlier_Flag == "Normal")
over_df_lin <- aug_lin |> filter(Outlier_Flag == "Overperformer")
under_df_lin <- aug_lin |> filter(Outlier_Flag == "Underperformer")

p_lin_resid <- ggplot() +

  annotate("rect",
    xmin = -Inf, xmax = Inf,
    ymin = underp_thr_lin, ymax = overp_thr_lin,
    fill = "#F5F5F5", alpha = 0.8) +

  geom_hline(yintercept = 0, colour = "grey20", linewidth = 0.9) +
  geom_hline(yintercept = overp_thr_lin, linetype = "dashed", colour = "#1D6FA4", linewidth = 0.75) +
  geom_hline(yintercept = underp_thr_lin, linetype = "dashed", colour = "#C0392B", linewidth = 0.75) +

  geom_point(data = bg_df_lin, aes(x = Fitted_PPG, y = Residual_PPG),
    colour = "grey72", size = 1.5, alpha = 0.55) +

  geom_point(data = under_df_lin, aes(x = Fitted_PPG, y = Residual_PPG),
    colour = "#C0392B", fill = "#C0392B", shape = 25, size = 2.2, alpha = 0.6) +

  geom_point(data = over_df_lin, aes(x = Fitted_PPG, y = Residual_PPG),
    colour = "#1D6FA4", fill = "#1D6FA4", shape = 24, size = 2.4, alpha = 0.7) +

  geom_point(data = label_df_lin, aes(x = Fitted_PPG, y = Residual_PPG),
    colour = "#F39C12", fill = "#F39C12", shape = 23, size = 4.5) +

  geom_label_repel(
    data = label_df_lin,
    aes(x = Fitted_PPG, y = Residual_PPG, label = Label),
    fill = "white",
    colour = "#2C3E50",
    size = 3.2,
    fontface = "bold",
    label.size = 0.3,
    label.r = unit(0.2, "lines"),
    box.padding = 0.6,
    point.padding = 0.4,

```

```

    segment.colour = "#F39C12",
    segment.size   = 0.5,
    min.segment.length = 0.2,
    max.overlaps   = 20
  ) +

  annotate("text", x = max(aug_lin$Fitted_PPG) - 0.15, y = overp_thr_lin + 0.09,
            label = "Overperformer zone (above +1.5 SD)", colour = "#1D6FA4",
            size = 3, hjust = 1, fontface = "italic") +
  annotate("text", x = min(aug_lin$Fitted_PPG) + 0.05, y = underp_thr_lin - 0.09,
            label = "Underperformer zone (below -1.5 SD)", colour = "#C0392B",
            size = 3, hjust = 0, fontface = "italic") +

  labs(
    title   = "Residuals vs Fitted - Linear PPG Model",
    subtitle = sprintf(
      "N = %d • Residual SD = %.3f PPG • Orange diamonds = key story clubs (labelled where still no",
      nrow(aug_lin), resid_sd_lin
    ),
    x       = "Fitted Points Per Game (linear model prediction)",
    y       = "Residual (Actual - Predicted)\nPositive = performed BETTER than predicted",
    caption = "Model: Points_Per_Game ~ Financial_Rank (no polynomial term, per Jacob's feedback)"
  ) +
  theme_clean

path_lin_resid <- file.path(OUTPUT_DIR, "linear_ppg_residuals.png")
ggsave(path_lin_resid, p_lin_resid, width = 10, height = 7, dpi = 150)
track(path_lin_resid)
cat(sprintf(" Saved: %s\n", path_lin_resid))
cat("TASK 4 complete.\n\n")

#
# TASK 5 - GAM fitted smooth vs Financial_Rank with CI band
# → gam_smooth_confidence.png
#
cat("=== TASK 5: GAM Fitted Smooth with Confidence Band ===\n")

fr_range <- range(model_df_ppg$Financial_Rank, na.rm = TRUE)
fr_seq   <- seq(fr_range[1], fr_range[2], length.out = 300)

gam_pred <- predict(gam_ppg, newdata = data.frame(Financial_Rank = fr_seq),
                    se.fit = TRUE)

gam_smooth_df <- tibble(
  Financial_Rank = fr_seq,
  fit            = gam_pred$fit,
  se            = gam_pred$se.fit
) |>
  mutate(lwr = fit - 1.96 * se, upr = fit + 1.96 * se)

# Where the CI band is widest (least data-supported) vs narrowest
widest_pt   <- gam_smooth_df |> slice_max(order_by = upr - lwr, n = 1)
narrowest_pt <- gam_smooth_df |> slice_min(order_by = upr - lwr, n = 1)

```

```

cat(sprintf(" CI band narrowest (best-supported) near FR = %.1f (width = %.3f PPG)\n",
  narrowest_pt$Financial_Rank, narrowest_pt$upr - narrowest_pt$lwr))
cat(sprintf(" CI band widest (least-supported) near FR = %.1f (width = %.3f PPG)\n\n",
  widest_pt$Financial_Rank, widest_pt$upr - widest_pt$lwr))

p_gam_smooth <- ggplot() +
  geom_jitter(data = model_df_ppg, aes(x = Financial_Rank, y = Points_Per_Game, colour = League),
    alpha = 0.3, size = 1.1, width = 0.18, height = 0.02) +
  geom_ribbon(data = gam_smooth_df, aes(x = Financial_Rank, ymin = lwr, ymax = upr),
    fill = "#2C3E73", alpha = 0.18) +
  geom_line(data = gam_smooth_df, aes(x = Financial_Rank, y = fit),
    colour = "#2C3E73", linewidth = 1.2) +
  annotate("label", x = fr_range[1] + 0.05 * diff(fr_range),
    y = max(gam_smooth_df$upr) - 0.04 * diff(range(model_df_ppg$Points_Per_Game, na.rm = TRUE)),
    label = sprintf("Deviance explained = %.3f\nedf = %.2f", gam_devexpl, gam_edf),
    colour = "#2C3E73", fill = "white", size = 3.6, fontface = "bold", hjust = 0) +
  scale_colour_manual(values = league_colours) +
  scale_x_continuous(breaks = c(1, 5, 10, 15, 20)) +
  labs(
    title = "GAM Fitted Smooth - Points Per Game vs Financial Rank",
    subtitle = "Shaded band = 95% confidence interval; widens where financial-rank values are sparser",
    x = "Financial Rank (FR 1 = richest squad)",
    y = "Points Per Game",
    colour = "League",
    caption = "Model: Points_Per_Game ~ s(Financial_Rank), fit with mgcv::gam(method = \"REML\")"
  ) +
  theme_clean

path_gam_smooth <- file.path(OUTPUT_DIR, "gam_smooth_confidence.png")
ggsave(path_gam_smooth, p_gam_smooth, width = 10, height = 7, dpi = 150)
track(path_gam_smooth)
cat(sprintf(" Saved: %s\n", path_gam_smooth))
cat("TASK 5 complete.\n\n")

#
# TASK 6 - GAM residuals vs fitted (same visual style as E5 /
# linear_ppg_residuals.png), plus a check for any new
# standout outliers that emerge specifically under the GAM
# + gam_ppg_residuals.png
#
cat("=== TASK 6: GAM Residuals vs Fitted ===\n")

aug_gam <- augment(gam_ppg, data = model_df_ppg) |>
  rename(Fitted_PPG = .fitted, Residual_PPG = .resid)

resid_sd_gam <- sd(aug_gam$Residual_PPG, na.rm = TRUE)
overp_thr_gam <- 1.5 * resid_sd_gam
underp_thr_gam <- -1.5 * resid_sd_gam

aug_gam <- aug_gam |>
  mutate(
    Outlier_Flag = case_when(

```



```

    Residual_PPG > overp_thr_gam ~ "Overperformer",
    Residual_PPG < underp_thr_gam ~ "Underperformer",
    TRUE ~ "Normal"
  )
)

cat(sprintf(" Residual SD: %.3f PPG | Overp. threshold: > %.3f | Underp. threshold: < %.3f\n",
            resid_sd_gam, overp_thr_gam, underp_thr_gam))
cat(sprintf(" For comparison - residual SD: linear PPG = %.3f PPG | quadratic Model H (PPG) = 0.254\n",
            resid_sd_lin))

# Story clubs already flagged in earlier work (E5's PPG-model plot and
# this script's linear-model plot) - used below to detect whether the
# GAM surfaces any *new* standout club-seasons, not just the familiar ones.
previously_flagged <- tibble::tribble(
  ~Team_Pattern, ~Season,
  "Leicester", "2015-2016",
  "Leverkusen", "2023-2024",
  "Lille", "2020-2021",
  "Napoli", "2024-2025",
  "Liverpool", "2024-2025"
)

is_previously_flagged <- function(team, season) {
  any(mapply(function(p, s) grepl(p, team) & season == s,
    previously_flagged$Team_Pattern, previously_flagged$Season))
}

aug_gam <- aug_gam |>
  rowwise() |>
  mutate(Previously_Flagged = is_previously_flagged(Team, Season)) |>
  ungroup()

leicester_gam <- aug_gam |> filter(grepl("Leicester", Team), Season == "2015-2016")
leverkusen_gam <- aug_gam |> filter(grepl("Leverkusen", Team), Season == "2023-2024")
if (nrow(leicester_gam) == 1) {
  cat(sprintf(" Leicester City 2015-16 under the GAM: Residual = %+3f PPG (%s)\n",
    leicester_gam$Residual_PPG, leicester_gam$Outlier_Flag))
}
if (nrow(leverkusen_gam) == 1) {
  cat(sprintf(" Bayer Leverkusen 2023-24 under the GAM: Residual = %+3f PPG (%s)\n",
    leverkusen_gam$Residual_PPG, leverkusen_gam$Outlier_Flag))
}

new_over <- aug_gam |>
  filter(Outlier_Flag == "Overperformer", !Previously_Flagged) |>
  arrange(desc(Residual_PPG)) |> slice(1)
new_under <- aug_gam |>
  filter(Outlier_Flag == "Underperformer", !Previously_Flagged) |>
  arrange(Residual_PPG) |> slice(1)

if (nrow(new_over) == 1) {
  cat(sprintf(" New standout overperformer under the GAM (not previously flagged): %s %s (Residual = %s)\n",
    new_over$Team, new_over$Season, new_over$Residual_PPG))
}

```

```

if (nrow(new_under) == 1) {
  cat(sprintf(" New standout underperformer under the GAM (not previously flagged): %s %s (Residual =
              new_under$Team, new_under$Season, new_under$Residual_PPG))
}
cat("\n")

label_df_gam <- bind_rows(
  leicester_gam |> filter(Outlier_Flag != "Normal"),
  leverkusen_gam |> filter(Outlier_Flag != "Normal"),
  new_over,
  new_under
) |>
  distinct(Team, Season, .keep_all = TRUE) |>
  mutate(Label = paste0(Team, "\n", Season))

bg_df_gam <- aug_gam |> filter(Outlier_Flag == "Normal")
over_df_gam <- aug_gam |> filter(Outlier_Flag == "Overperformer")
under_df_gam <- aug_gam |> filter(Outlier_Flag == "Underperformer")

p_gam_resid <- ggplot() +

  annotate("rect",
    xmin = -Inf, xmax = Inf,
    ymin = underp_thr_gam, ymax = overp_thr_gam,
    fill = "#F5F5F5", alpha = 0.8) +

  geom_hline(yintercept = 0, colour = "grey20", linewidth = 0.9) +
  geom_hline(yintercept = overp_thr_gam, linetype = "dashed", colour = "#1D6FA4", linewidth = 0.75) +
  geom_hline(yintercept = underp_thr_gam, linetype = "dashed", colour = "#C0392B", linewidth = 0.75) +

  geom_point(data = bg_df_gam, aes(x = Fitted_PPG, y = Residual_PPG),
    colour = "grey72", size = 1.5, alpha = 0.55) +

  geom_point(data = under_df_gam, aes(x = Fitted_PPG, y = Residual_PPG),
    colour = "#C0392B", fill = "#C0392B", shape = 25, size = 2.2, alpha = 0.6) +

  geom_point(data = over_df_gam, aes(x = Fitted_PPG, y = Residual_PPG),
    colour = "#1D6FA4", fill = "#1D6FA4", shape = 24, size = 2.4, alpha = 0.7) +

  geom_point(data = label_df_gam, aes(x = Fitted_PPG, y = Residual_PPG),
    colour = "#F39C12", fill = "#F39C12", shape = 23, size = 4.5) +

  geom_label_repel(
    data = label_df_gam,
    aes(x = Fitted_PPG, y = Residual_PPG, label = Label),
    fill = "white",
    colour = "#2C3E50",
    size = 3.2,
    fontface = "bold",
    label.size = 0.3,
    label.r = unit(0.2, "lines"),
    box.padding = 0.6,
    point.padding = 0.4,

```

```

    segment.colour = "#F39C12",
    segment.size   = 0.5,
    min.segment.length = 0.2,
    max.overlaps   = 20
  ) +

  annotate("text", x = max(aug_gam$Fitted_PPG) - 0.15, y = overp_thr_gam + 0.09,
    label = "Overperformer zone (above +1.5 SD)", colour = "#1D6FA4",
    size = 3, hjust = 1, fontface = "italic") +
  annotate("text", x = min(aug_gam$Fitted_PPG) + 0.05, y = underp_thr_gam - 0.09,
    label = "Underperformer zone (below -1.5 SD)", colour = "#C0392B",
    size = 3, hjust = 0, fontface = "italic") +

  labs(
    title   = "Residuals vs Fitted - GAM (PPG) Model",
    subtitle = sprintf(
      "N = %d • Residual SD = %.3f PPG (linear = 0.271, quadratic = 0.254) • Orange diamonds = story",
      nrow(aug_gam), resid_sd_gam
    ),
    x       = "Fitted Points Per Game (GAM prediction)",
    y       = "Residual (Actual - Predicted)\nPositive = performed BETTER than predicted",
    caption = "Model: Points_Per_Game ~ s(Financial_Rank), fit with mgcv::gam(method = \"REML\")"
  ) +
  theme_clean

path_gam_resid <- file.path(OUTPUT_DIR, "gam_ppg_residuals.png")
ggsave(path_gam_resid, p_gam_resid, width = 10, height = 7, dpi = 150)
track(path_gam_resid)
cat(sprintf(" Saved: %s\n", path_gam_resid))
cat("TASK 6 complete.\n\n")

#
# PHASE 6: Output Inventory
#
cat("=== PHASE 6: Output Inventory ===\n\n")
cat(sprintf(" Script produced %d output file(s) in %s/:\n", length(OUTPUT_FILES), OUTPUT_DIR))
for (f in OUTPUT_FILES) {
  size_kb <- round(file.info(f)$size / 1024, 1)
  cat(sprintf(" %-65s %6.1f KB\n", f, size_kb))
}
cat("\n16_linear_gam_revision.R complete.\n")

```

Report generated with R Markdown. Data sources: ESPN FC (league standings), Transfermarkt (squad market values). All analysis scripts are available in the project repository. For questions, contact: mdmuktadir0611@gmail.com